
Year 8
Cells
Gamified
Mr. Barlow

Preface

The rate of technological change we are experiencing is unprecedented in human history. Science has and will continue to play a vital role in this advancement of our species.

Scientists are simply curious people so I encourage you, as you learn and in life, to constantly question, seek, source and explore new ideas.

Enjoy learning about Science!

Cover image: Bee (*Amegilla Asaropoda*). Bees can be distinguished from flies in that bees have four wings, where flies have only two. Bees collect pollen, from flowers, to feed their young. Wasps and flies do not do this.

All experiments must be performed under teacher direction in a school science laboratory. Teachers must ensure all appropriate safety precautions are taken.



Section 1

How do I play this game?

Welcome to your quest!

The aim of this game is simple ... collect as many stars as you can to progress to the highest level in the game - ‘Wizard’!

To progress up the levels you will need to complete various tasks or activities throughout this book.

There are two types of stars in this book ...

The first star is full, wherever you see this star it indicates that the task is compulsory and there is no option but to complete the task to get the star. There are a number of compulsory tasks.



The second type of star contains a number 6, when you see this star it means you have a choice. There are more than six of these stars in the book. You can choose to complete any of the tasks that have this star next to them but you can only get a maximum of six stars. So you only need to complete six of these tasks in total.



REMEMBER - AT ANY TIME YOUR TEACHER MAY ALLOCATE STARS FOR OTHER TASKS OR VARY WHICH TASKS YOU ARE ABLE TO ATTAIN A STAR FOR.

The more stars you are able to collect the further you will progress in the game. There are six different levels, the lowest level is ‘Recruit’ and to finish the game you need to collect all of the available stars in the entire book to achieve the rank of ‘Wizard’.

PLEASE NOTE:

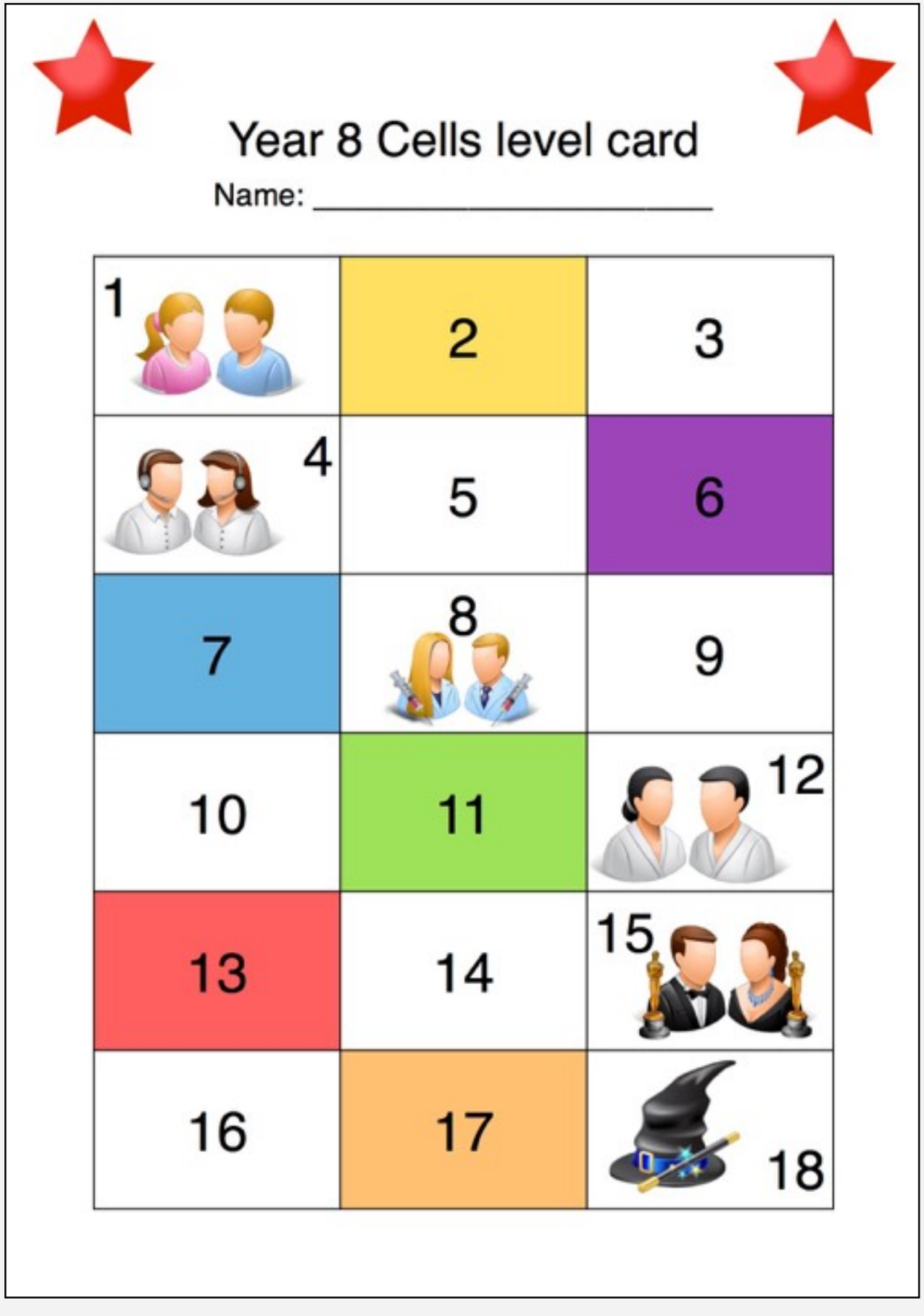
- to gain any single star the task has to be fully complete and done to the best of your ability.

- to finish the game you must complete each and every task associated with a star. You cannot achieve a rank of 'Wizard' unless you have gained all 18 available stars !!!

Good luck and start your learning.

Stars required to level-up in Year 8 Cells

LEVEL	LEVEL NAME	STARS REQUIRED
1	Recruit	1
2	Technician	4
3	Researcher	8
4	Master	12
5	Professor	15
6	**Wizard**	**18**



Level Card

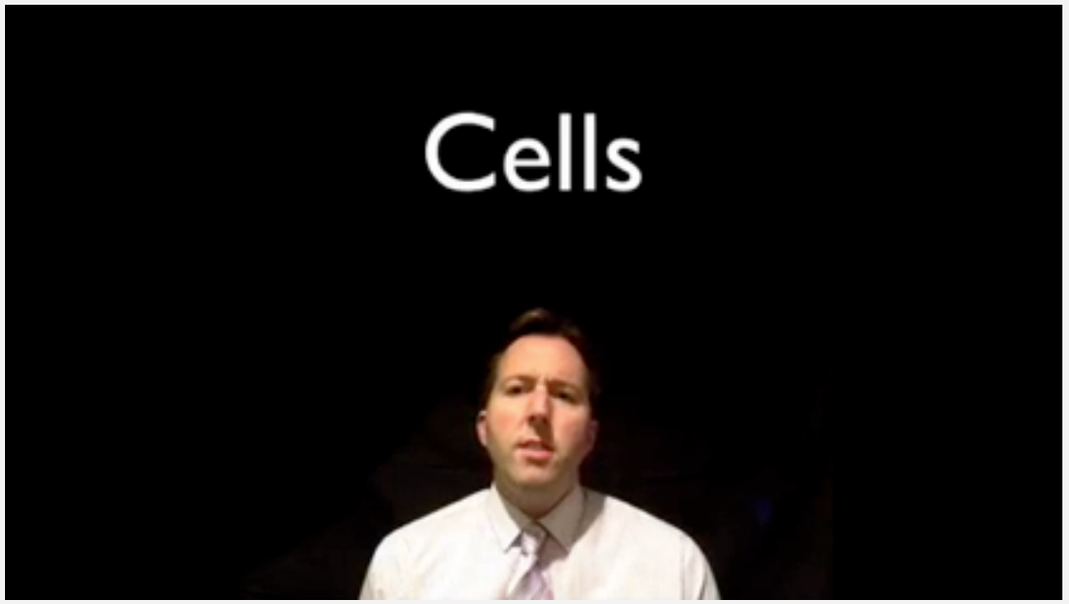
This card may be printed off and marked off as you progress through the game (capture a screen shot of this image to print it).

Chapter 1

Cells

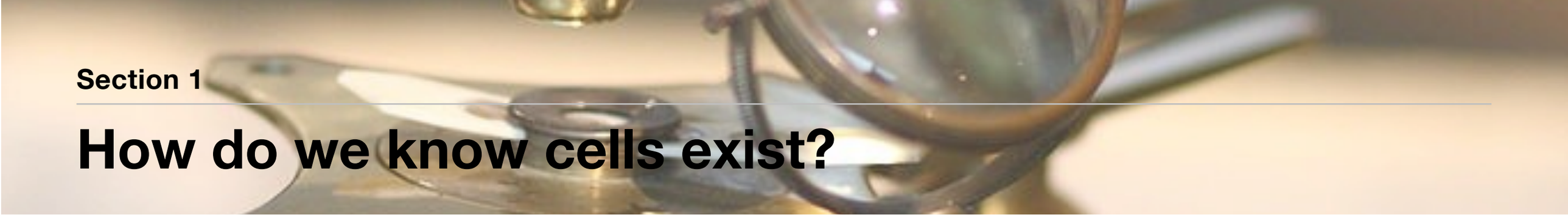
Cells are the basic units of living things and have specialised structures and functions.

Movie 1.1 A brief introduction to cells



At the end of this chapter you will be able to:

- examine a variety of cells using a light microscope, by digital technology or by viewing a simulation.
- distinguish plant cells from animal or fungal cells.
- identify structures within cells and describe their function.
- recognise that some organisms consist of a single cell.
- examine the specialised cells and tissues involved in structure and function of particular organs.
- recognise that cells reproduce via cell division.
- describe mitosis as cell division for growth and repair.
- distinguish between asexual and sexual reproduction.



Section 1

How do we know cells exist?

We now know that all living things are made up of **cells**. But the first time the word cell was used to refer to these tiny units of life was in 1665 by a British scientist named Robert Hooke. Hooke was one of the earliest scientists to study living things under a **microscope**. The microscopes of his day were not very strong, but Hooke was still able to make an important discovery. When he looked at a thin slice of cork under his microscope, he was surprised to see what looked like a honeycomb.

Soon after Robert Hooke discovered cells in cork, Anton van Leeuwenhoek (often called the "father of microscopy") in Holland made other important discoveries using a microscope. Leeuwenhoek made his own microscope lenses, and he was so good at it that his microscope was more powerful than other microscopes of his day. In fact, Leeuwenhoek's microscope was almost as strong as modern light microscopes. Using his microscope, Leeuwenhoek discovered tiny animals such as rotifers. Leeuwenhoek also discovered human blood cells. He even scraped plaque from his own teeth and observed it under the microscope. What do you think Leeuwenhoek saw in the plaque? He saw tiny living things with a single cell that he named

Image 1.1 Cell structure of Cork by Robert Hooke



The first ever drawing of cells was made by Robert Hooke in 1665.

Image 1.2 The microscope used by Robert Hooke in 1665



animalcules (“tiny animals”). Today, we call Leeuwenhoek’s animalcules bacteria.

For two centuries after the discoveries of Robert Hooke and Anton van Leeuwenhoek, biologists found cells everywhere. Biologists in the early part of the 19th century suggested that all living things were made of cells, but the role of cells as the primary building block of life was not discovered until 1839 when two German scientists, Theodor Schwann, a zoologist, and Matthias Jakob Schleiden, a botanist, suggested that cells were the basic unit of structure and function of all life. Later, in 1858, the German doctor Rudolf Virchow observed that cells divide to produce more cells. He proposed that all cells arise only from other cells. The collective observations of all three scientists form the Cell Theory, which states that:

- all organisms are made up of one or more cells,
- the cell is the smallest living organisational unit,
- all cells come from pre-existing cells.

The Cell Theory is one of the fundamental theories of biology.

Interactive 1.1 3D Modern Light Microscope



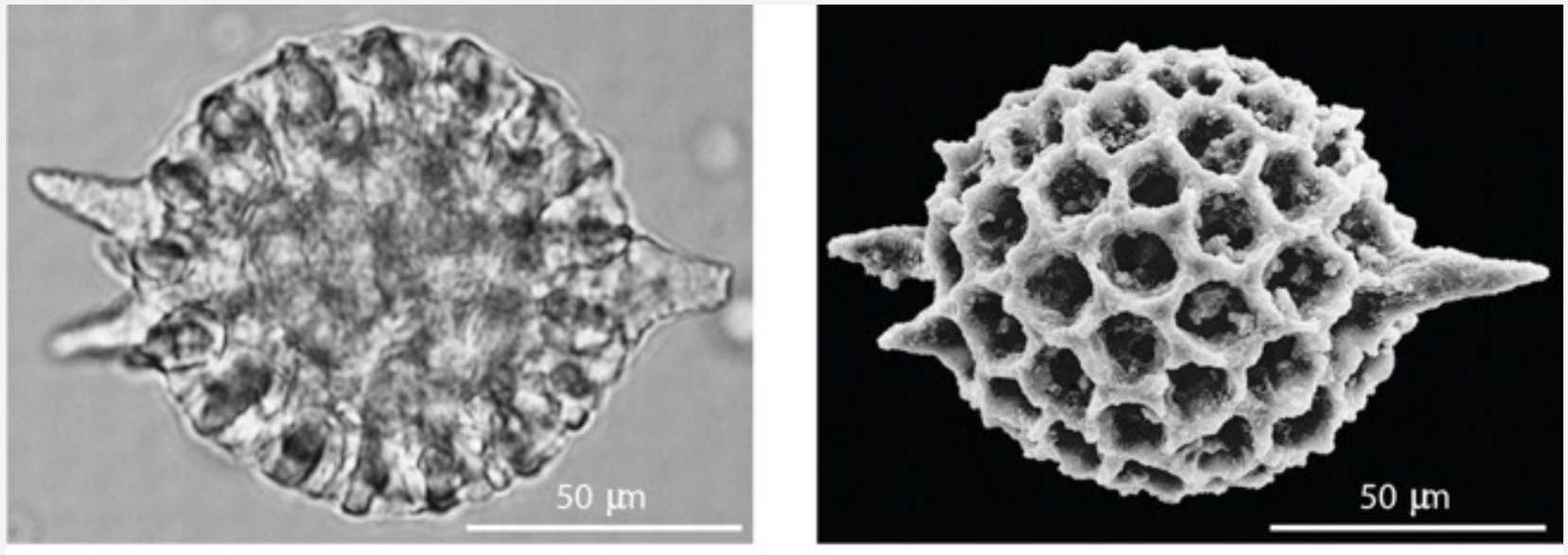
Explore this 3D microscope model. Can you recognise the various parts?

Of course our knowledge of cells has expanded greatly since the time of Hooke and Leeuwenhoek and so has the quality of the microscopes we use.

Today there are two main types of microscopes, **light microscopes** and **electron microscopes**.

Light microscopes are significantly cheaper than electron microscopes. Scientists are able to view specimens magnified over 1,000 times under a light microscope although electron microscopes can magnify specimens up to 1,000,000 times! So although electron microscopes are far more expensive they also

Image 1.3 A small organism called a Radiolarian as seen under a light and electron microscope



(a) Radiolarian under light microscope

(b) Radiolarian under electron microscope

Table 1.1 Comparison of Light and Electron Microscopes

Light Microscope	Up to x1,500 magnification	Can view specimens in natural colours	Live (or dead) and moving specimens can be seen	Cheap and simple to use	Some detail of the specimen can be seen
Electron Microscope	Up to x1,000,000 magnification	Only creates monochrome images (although they are often artificially colour later)	Can only view dead and still objects	Very expensive to buy and a high level of technical skills needed to use	Extremely high level of specimen detail is able to be seen

show far greater detail in the specimen being studied.

Light microscopes, which you may use, have many parts:

Ocular Lens: the lens at the top that you look through. It is usually 10x power.

Body Tube: Connects the ocular lens (or eyepiece) to the objective lenses.

Arm: Supports the tube and connects it to the base.

Base: The bottom of the microscope, used for support.

Light source: Illuminates the specimen.

Stage: The flat platform where you place your slides. Stage clips hold the slides in place.

Revolving Nosepiece: This is the part that holds two or more objective lenses and can be rotated to easily change power.

Objective Lenses: Usually you will find 3 or 4 objective lenses on a microscope. They almost always consist of 4x, 10x, 40x and 100x powers. When coupled with a 10x (most common) ocular lens, we get total magnifications of 40x (4x times 10x), 100x, 400x and 1000x.

Diaphragm or Iris: The diaphragm has different sized holes and is used to vary the intensity of light that is projected upward into the slide.

Adjustment knobs: The coarse and fine adjustment knobs are used to focus the specimen.

Image 1.4 The parts of a light microscope

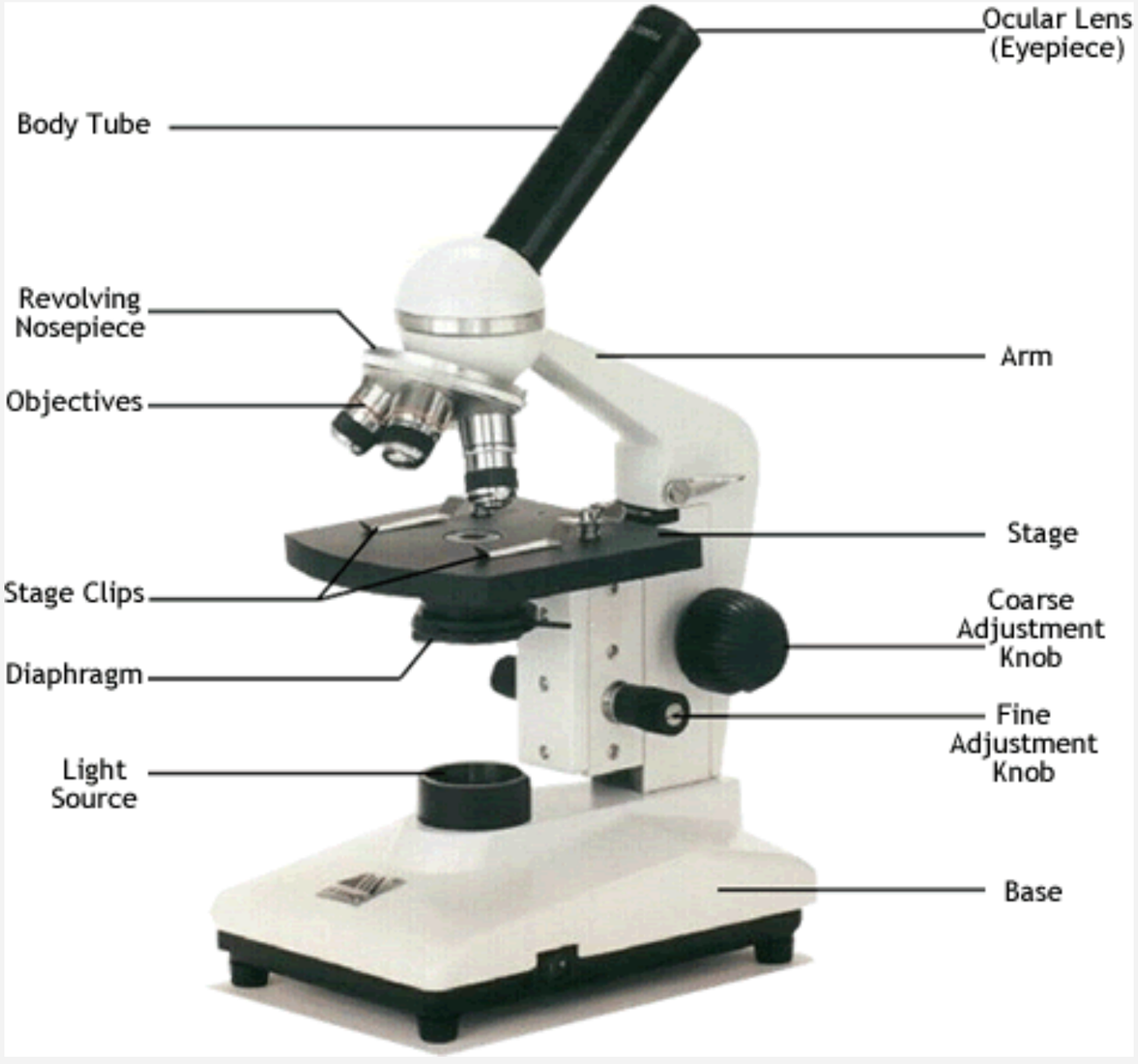


Image 1.5 A single red blood cell on the tip of a needle as seen using an electron microscope

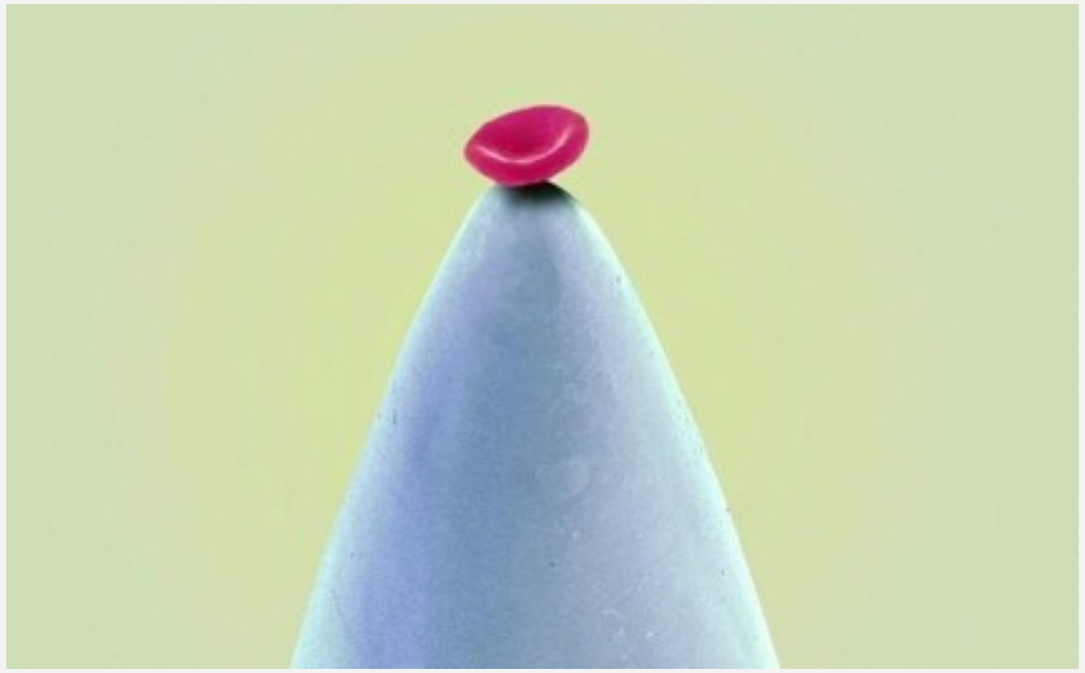
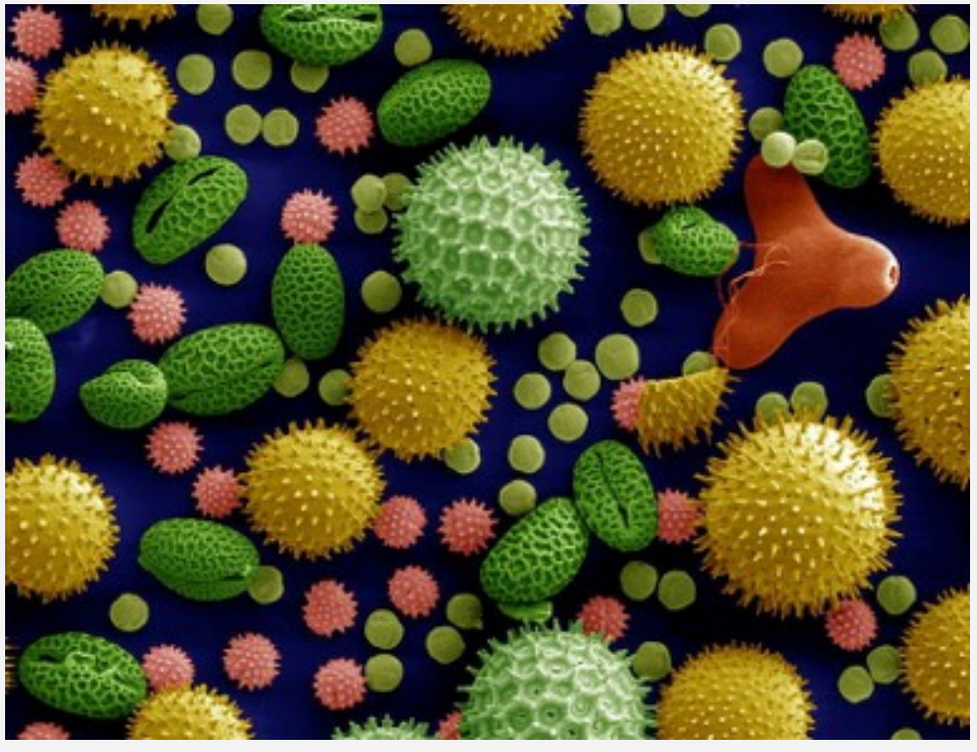


Image 1.6 Pollen from a variety of common plants as seen using an electron microscope (false colour added)



How to use a microscope

1. When moving your microscope, always carry it with both hands. Grasp the arm with one hand and place the other hand under the base for support.
2. Turn the revolving nosepiece so that the lowest power objective lens is "clicked" into position (this is also the shortest objective lens).
3. Your microscope slide should be prepared with a coverslip or cover glass over the specimen. Place the microscope slide on the stage and fasten it with the stage clips. You can push down on the back end of the stage clip to open it.
4. Look at the objective lens and the stage from the side and turn the coarse focus knob so that the objective lens moves downward. Move it as far as it will go without touching the slide so as not to shatter the slide!
5. Slowly turn the coarse adjustment so that the objective lens goes up (away from the slide). Continue until the image comes into focus. Use the fine adjustment, if available, for fine focusing.
6. Move the microscope slide around so that the image is in the centre of the field of view and re-adjust the light or diaphragm for the clearest image.

7. Now, you should be able to change to the next (more powerful) objective lenses with only minimal use of the focusing adjustment.

Microscope Troubleshooting

Occasionally you may have trouble with working your microscope. Here are some common problems and solutions.

1. Image is too dark!

Adjust the diaphragm and make sure your light is on.

2. There's a spot in my viewing field, even when I move the slide the spot stays in the same place!

Your lens is dirty. Use lens paper, and only lens paper to carefully clean the objective and ocular lens. The ocular lens can be removed to clean the inside.

3. I can't see anything under high power!

Remember to start with the lowest power lens, if you can't focus under low power, then you won't be able to focus anything under high power.

4. Only half of my viewing field is lit, it looks like there's a half-moon in there!

You probably don't have your objective lens fully clicked into place.

EXPERIMENT: USING A MICROSCOPE

[To be performed under teacher direction in a science laboratory]



Aim: To develop skills and confidence when using a microscope and to explore a variety of small objects.

Materials: Various pre-prepare microscope slides, light microscope.

Method:

1. Pick one of the pre-prepared microscope slides and get it into focus using the microscope.
2. Using pencil, draw a diagram of what you see. Make sure your diagram includes clear and accurate labels (and is large enough to be easy to understand). Diagrams should also have a title with the specimen name and magnification used.
3. Repeat steps 1 and 2 for as many specimens as you can.

Results:

Draw a diagram of each specimen you observed under a microscope.

Discussion:

- Explain what you observed during this experiment.
- Did anything surprise you during this experiment?
- Discuss anything you could have done better when performing this experiment.

Conclusion:

Sum up your experiment with reference to the aim.



Image 1.7 How big are cells?

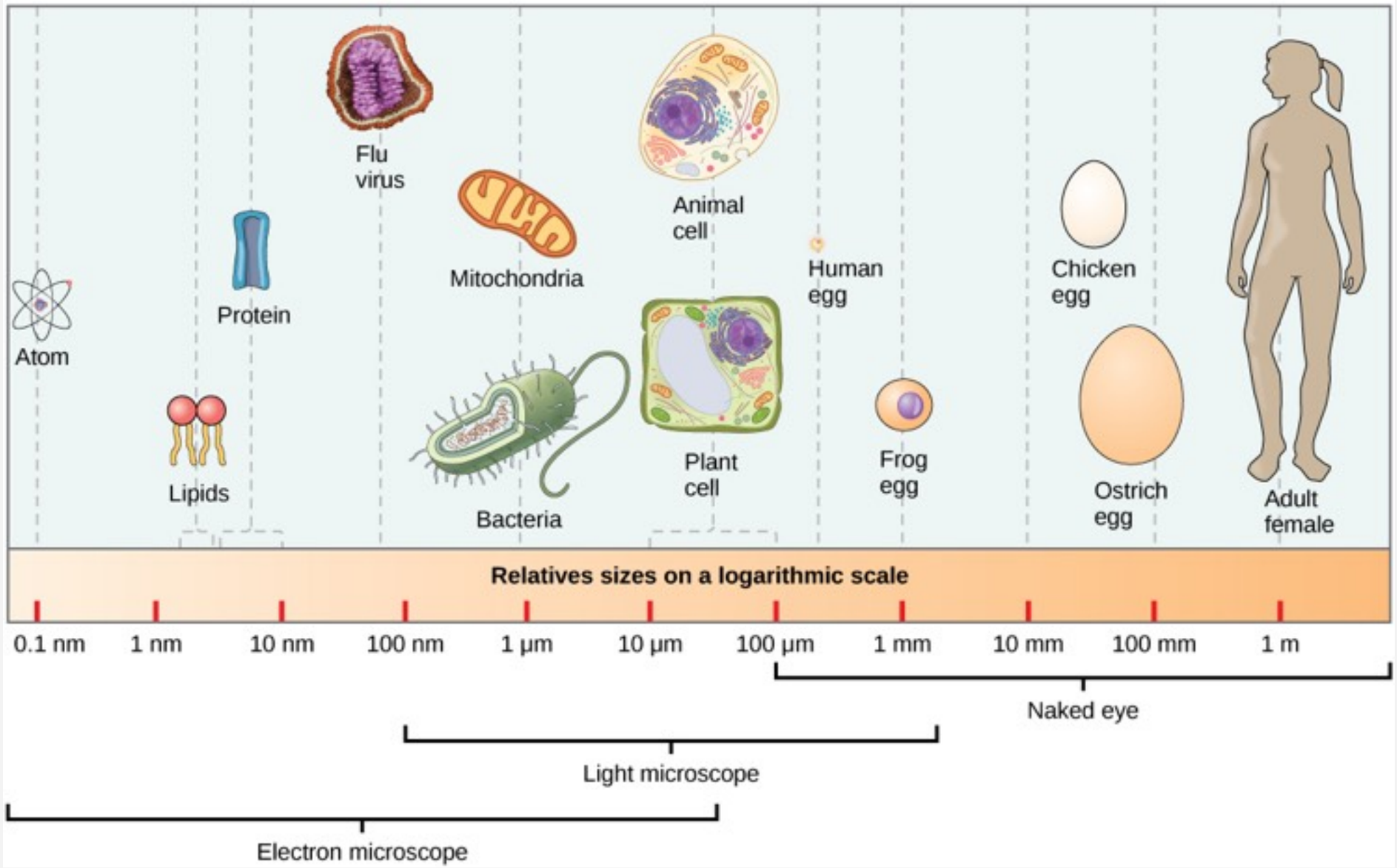
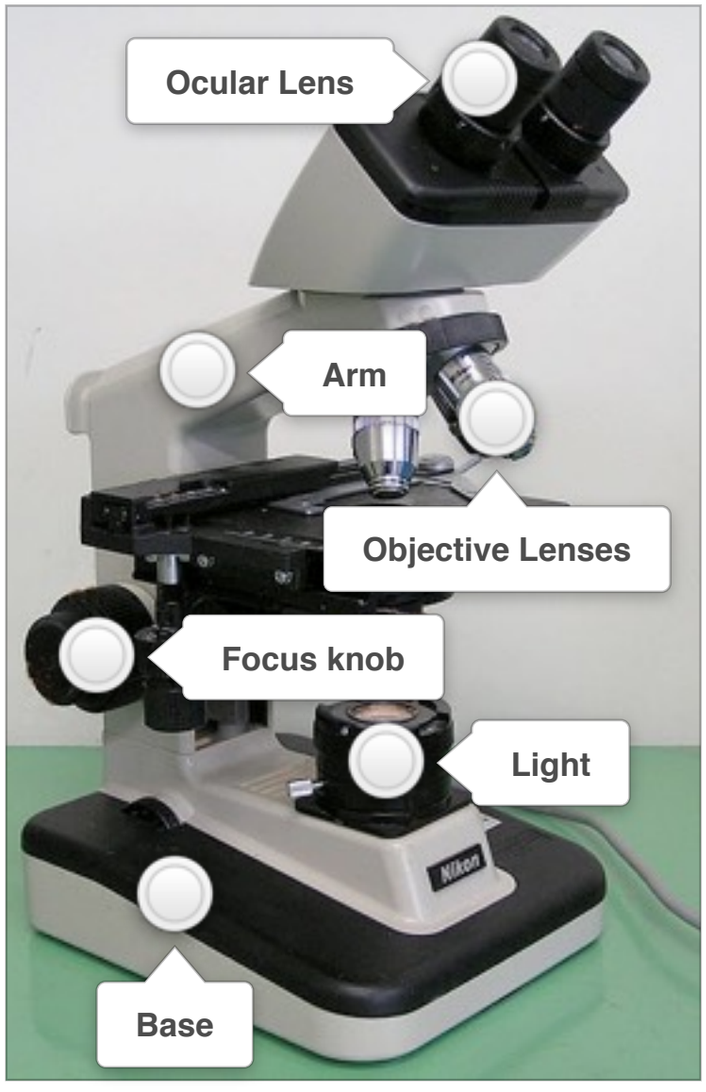


Table 1.2 How small is small?

Metre (m)	Millimetre (mm)	Micrometre (µm)	Nanometre
100 m	10^{-3} m	10^{-6} m	10^{-9} m
1 m	0.001 m	0.000001 m	0.000000001
	1/1000 m	1/1,000,000 m	1/1,000,000,000 m
	thousandth of a metre	millionth of a metre	billionth of a metre

Interactive 1.2 Parts of a microscope

Can you label this microscope?



Object… Ocular… Arm

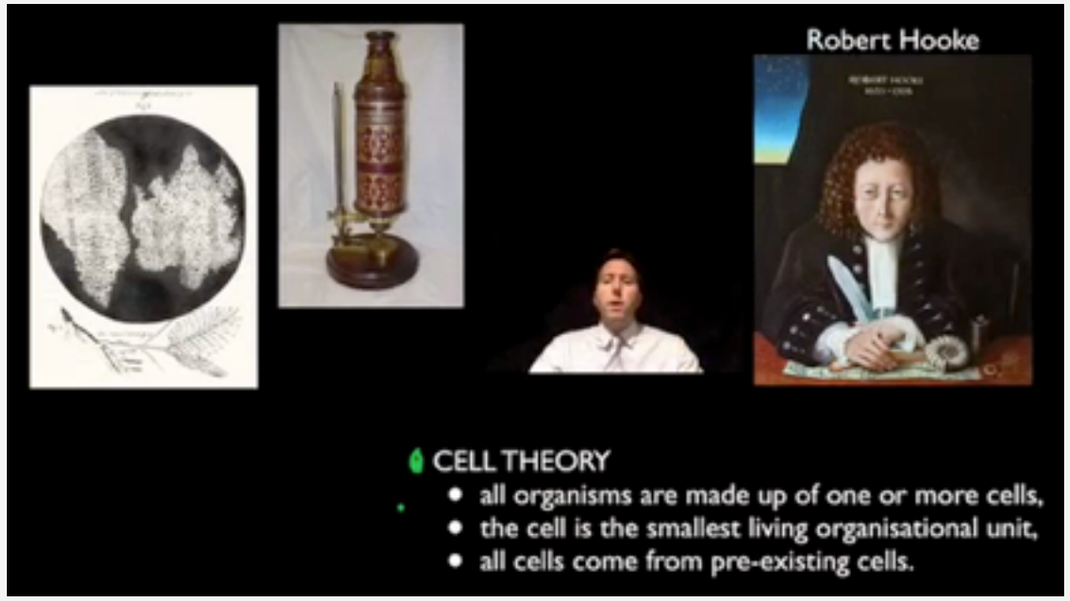
Focus … Base Light

Check Answer

QUESTIONS

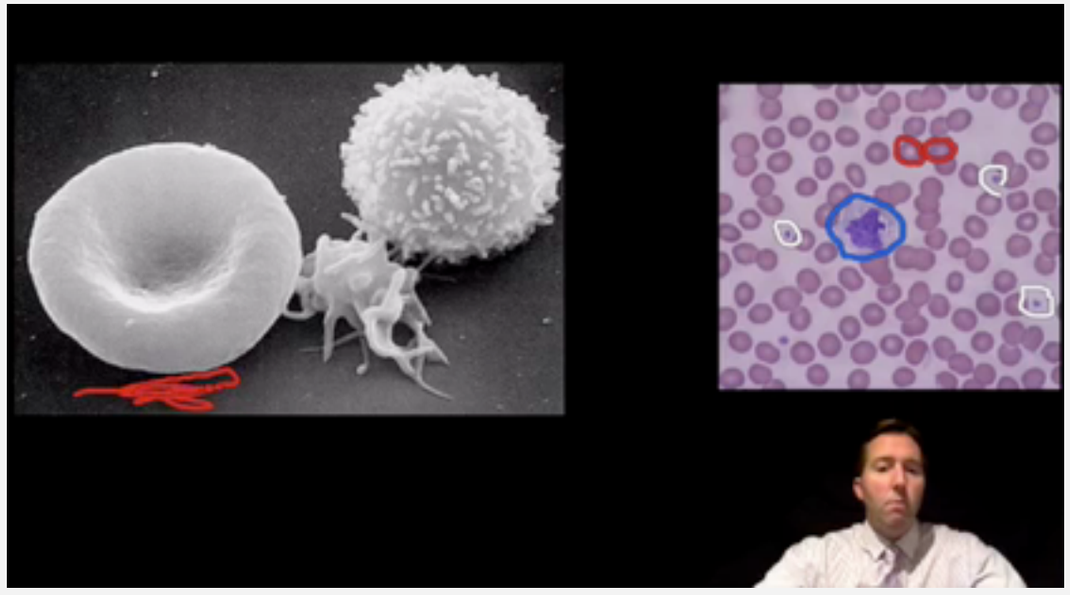
1. Re-write the following words in order from smallest to largest (you may need to research the meaning of the words first). Metre, yoctometre, millimetre, exametre, kilometre, attometre, micrometre, zettametre, terametre, femtometre, zeptometre, picometre, terametre, gigametre, nanometre, exametre, petametre, megametre, yottametre.
2. Try and calculate how tall you are in metres, millimetres and kilometres.
3. Create a venn diagram to compare a light microscope with an electron microscope.
4. Describe how our current understanding of biology may be different if the microscope had never ben invented.
5. If you had a chance to view one thing under an electron microscope, what would it be. Now research to see if it has been done!

Movie 1.2 How were cells discovered?



Using dot points, summarise the contents of this section.

Movie 1.3 Light and electron microscopes explained



Movie 1.4 What are the parts of a light microscope?

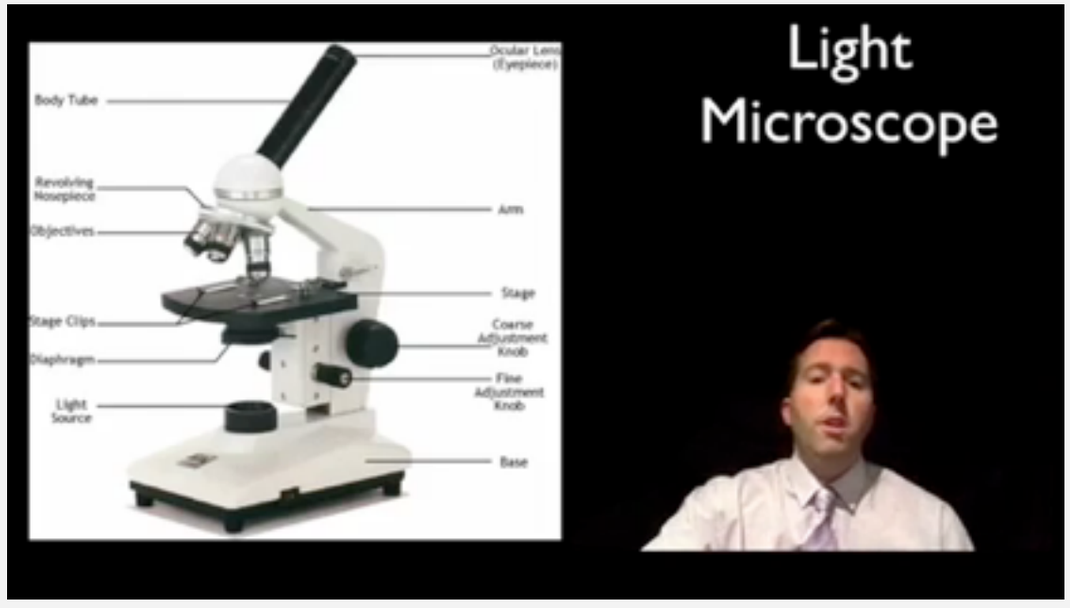
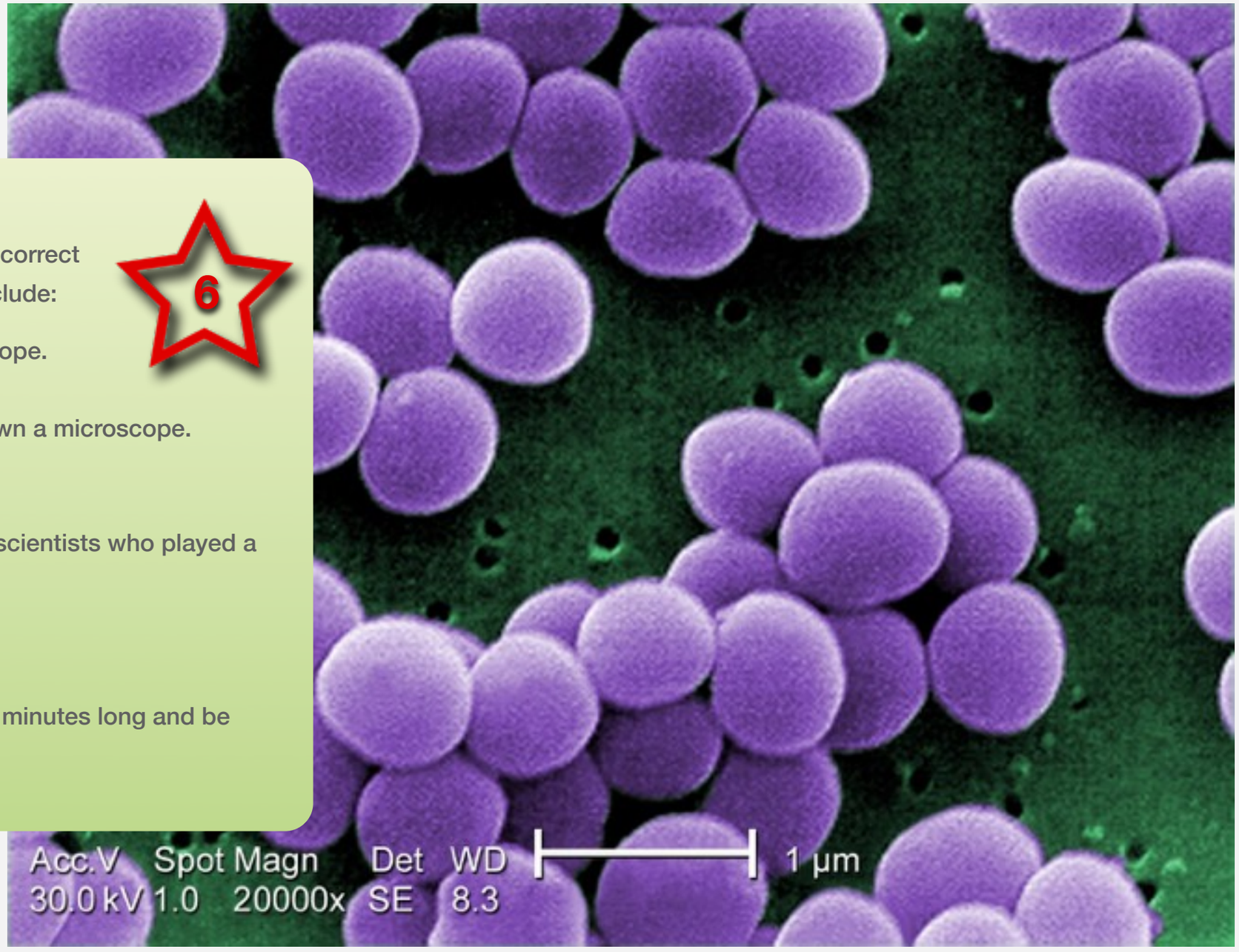


Image 1.8 Bacteria under an electron microscope



Under a very high magnification of 20,000x, this scanning electron micrograph (SEM) shows a strain of *Staphylococcus aureus* bacteria

PROJECT - Make a Microscope Movie

6

Create a short explanatory video about the correct use of a microscope. Your movie should include:

- Explanations about the parts of a microscope.
- Tips about how to use a microscope.
- Video or images of things you can see down a microscope.

You could also include:

- A brief history of the microscope and the scientists who played a role in its invention.
- Information about cells.
- Details of different types of microscopes.

Your video clip should be no more than 3-5 minutes long and be targeted at your peers.

Section 2

Are all cells the same?

All living things, or **organisms**, have several things in common, for example movement, growth and reproduction.

There are seven things that indicate an object is alive. The phrase **MRS GREN** is a way to remember them:

Table 1.3 What do all living things have in common?

M	**M**ovement	All living things move, even plants
R	(Cellular) **R**espiration	Getting energy from food (not 'breathing')
S	**S**ensitivity	Detecting changes in the surroundings
G	**G**rowth	All living things grow
R	**R**eproduction	Making more living things of the same type
E	**E**xcretion	Getting rid of waste
N	**N**utrition	Taking in and using food

All living things are also made from **cells** and all cells have certain parts in common. The common parts that cells share include a cell membrane, cytoplasm, ribosomes, and DNA.

1. The **cell membrane** (also called the plasma membrane) is a thin coat that surrounds a cell. It forms the physical boundary between the cell and its environment, so you can think of it as the "skin" of the cell.

2. **Cytoplasm** refers to all of the material inside the cell membrane, other than the nucleus. Cytoplasm is made up of a watery substance called cytosol, and contains other cell structures such as ribosomes.

3. **Ribosomes** are structures in the cytoplasm where proteins are made.

4. **DNA** (deoxyribonucleic acid) is a nucleic acid found in cells. It contains the genetic instructions for the cell.

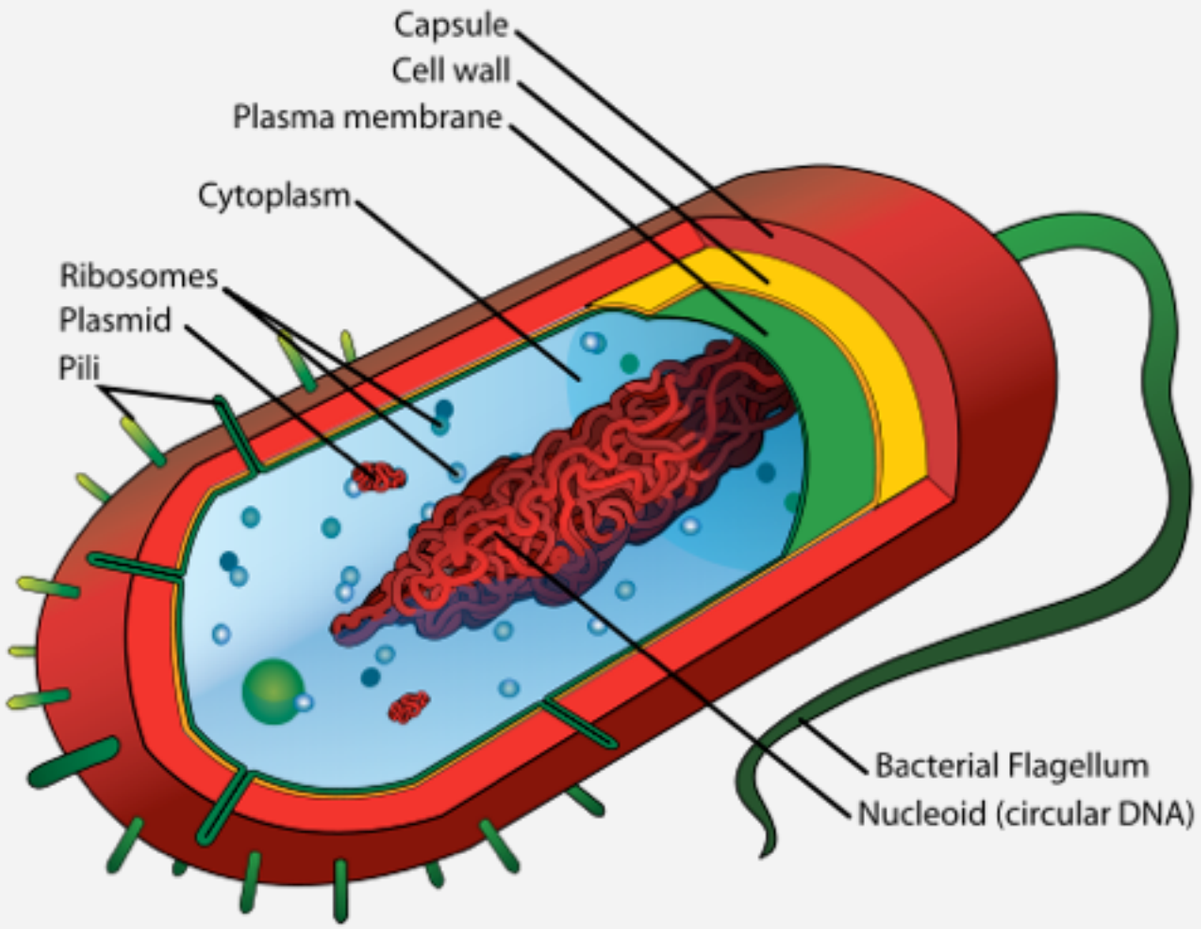
How did all known organisms come to have such similar cells? The similarities of all cells shows that all life on Earth has a common evolutionary history.

While all living things and all cells have some parts in common, there are actually many different types of cells.

The two main types of cells are prokaryotic cells and eukaryotic cells.

Prokaryotic cells are less complex than eukaryotic cells. Prokaryotic cells also lack a membrane-bound nucleus - they still have DNA but it is not contained within a membrane or 'skin' like structure. The prokaryotes are also **unicellular** organisms. This means that each living thing is actually made up of only one cell. Prokaryotic cells like bacteria are also much smaller than eukaryotic cells.

Image 1.9 Bacterial cell



Bacterial cells are prokaryotic as they do not have a membrane-bound nucleus.

Organisms whose cells do have a membrane-bound nucleus are called eukaryotes. **Eukaryotic cells** also contain other structures called **organelles** which are enclosed within membranes. As a result of the organelles found inside eukaryotes, these cells are considerably more complex than prokaryotic cells. All **multicellular** organisms are eukaryotes, including all animals and plants. A multicellular organism is one that is made up of many cells.

Image 1.10 The size difference between eukaryotes, prokaryotes and even smaller things.

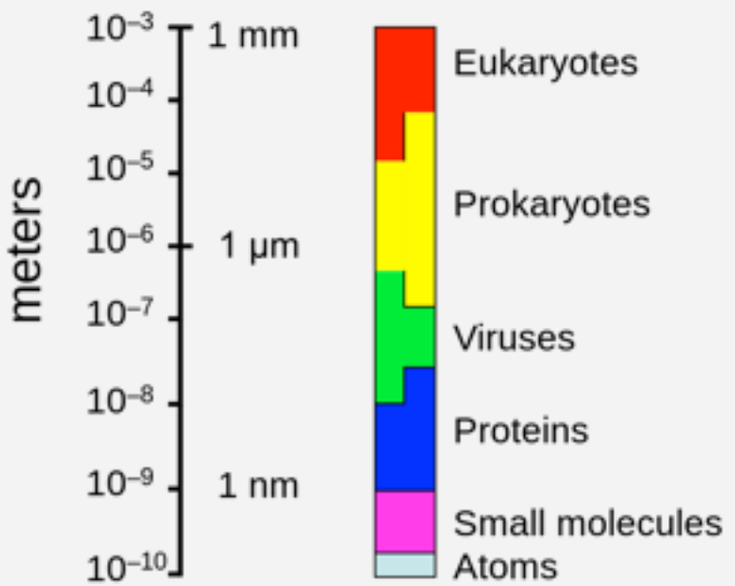
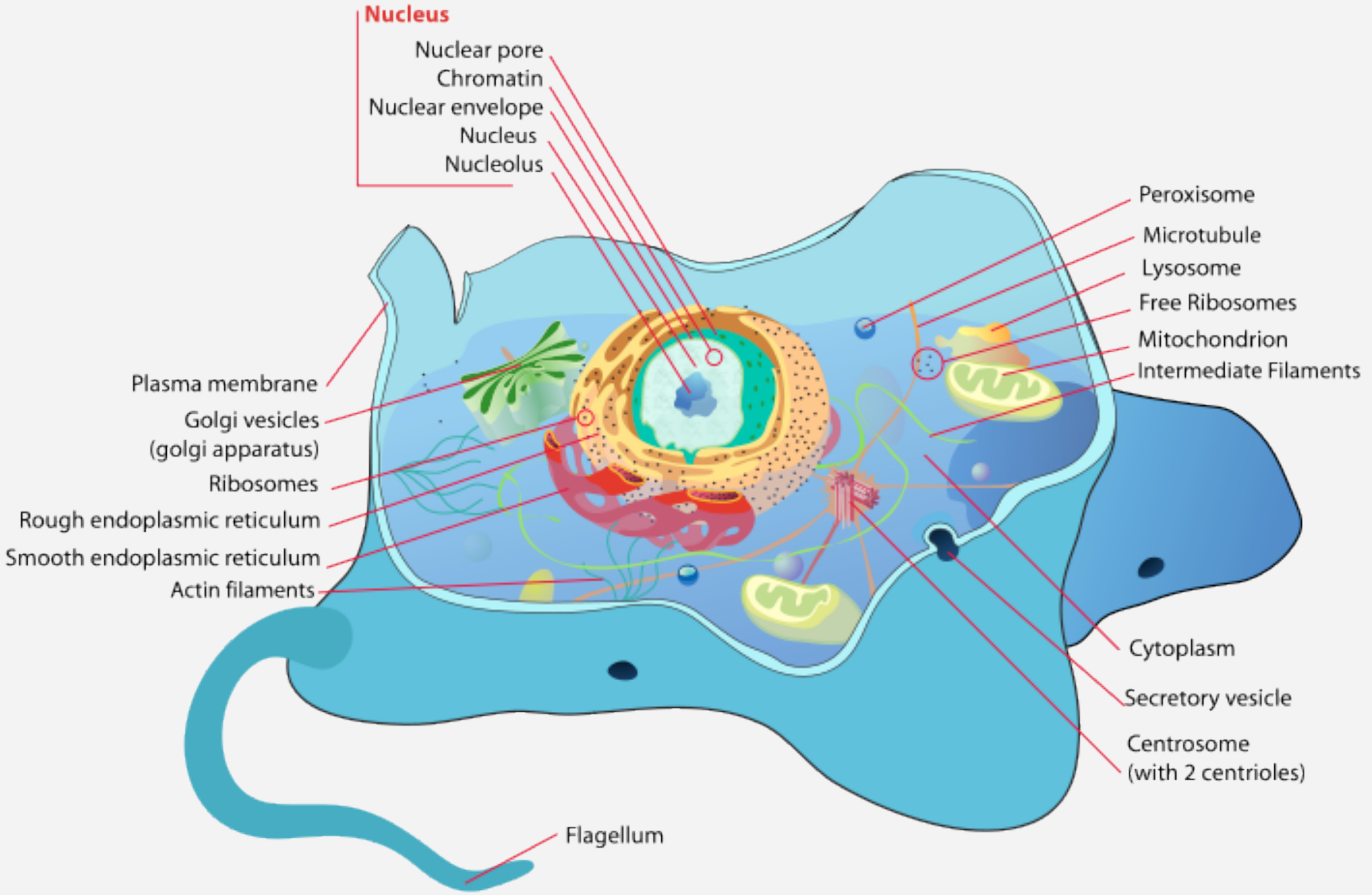
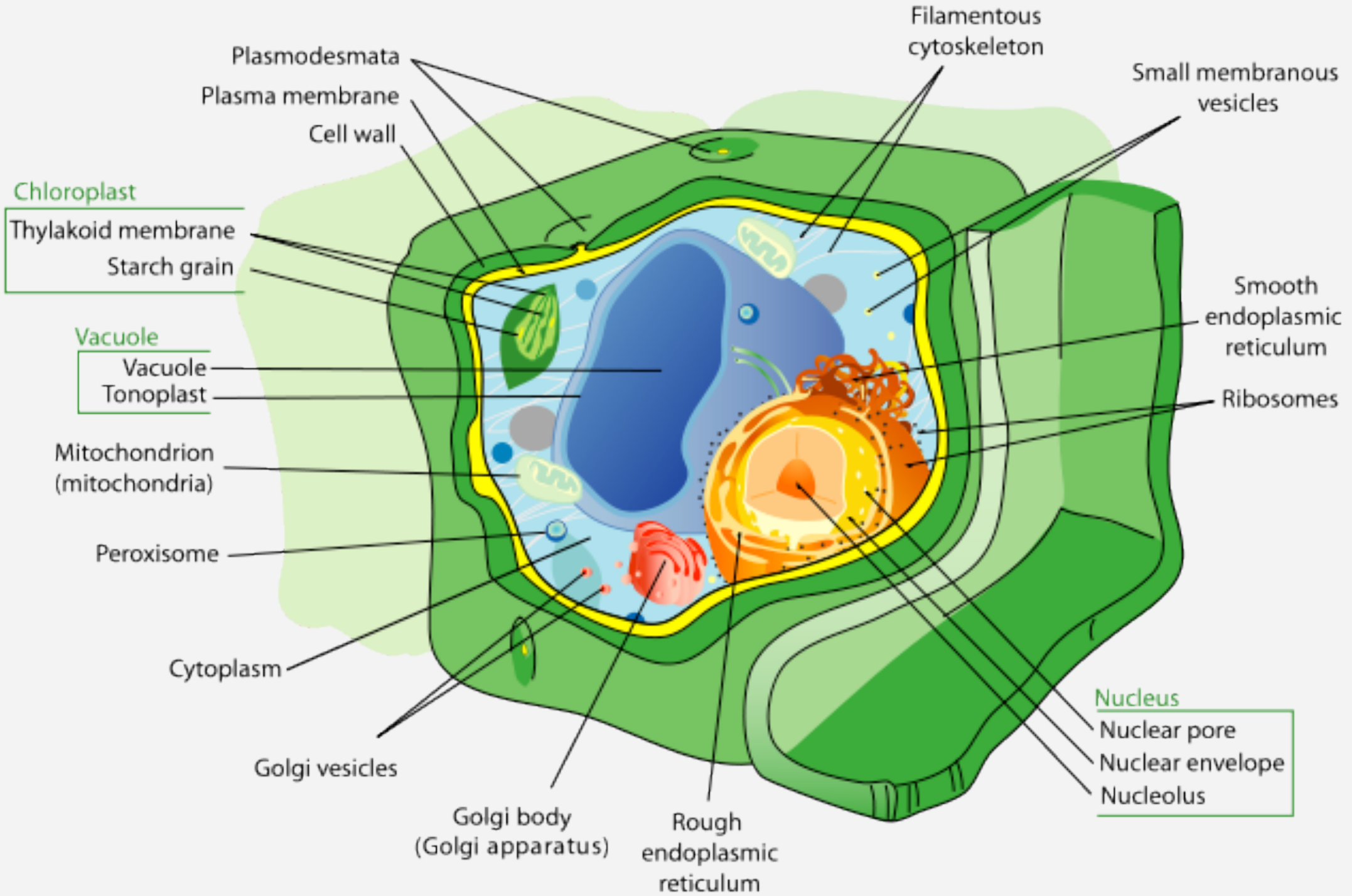


Image 1.11 Animal cell



Animal cells are eukaryotic.

Image 1.12 Plant cell



Plant cells are eukaryotic.

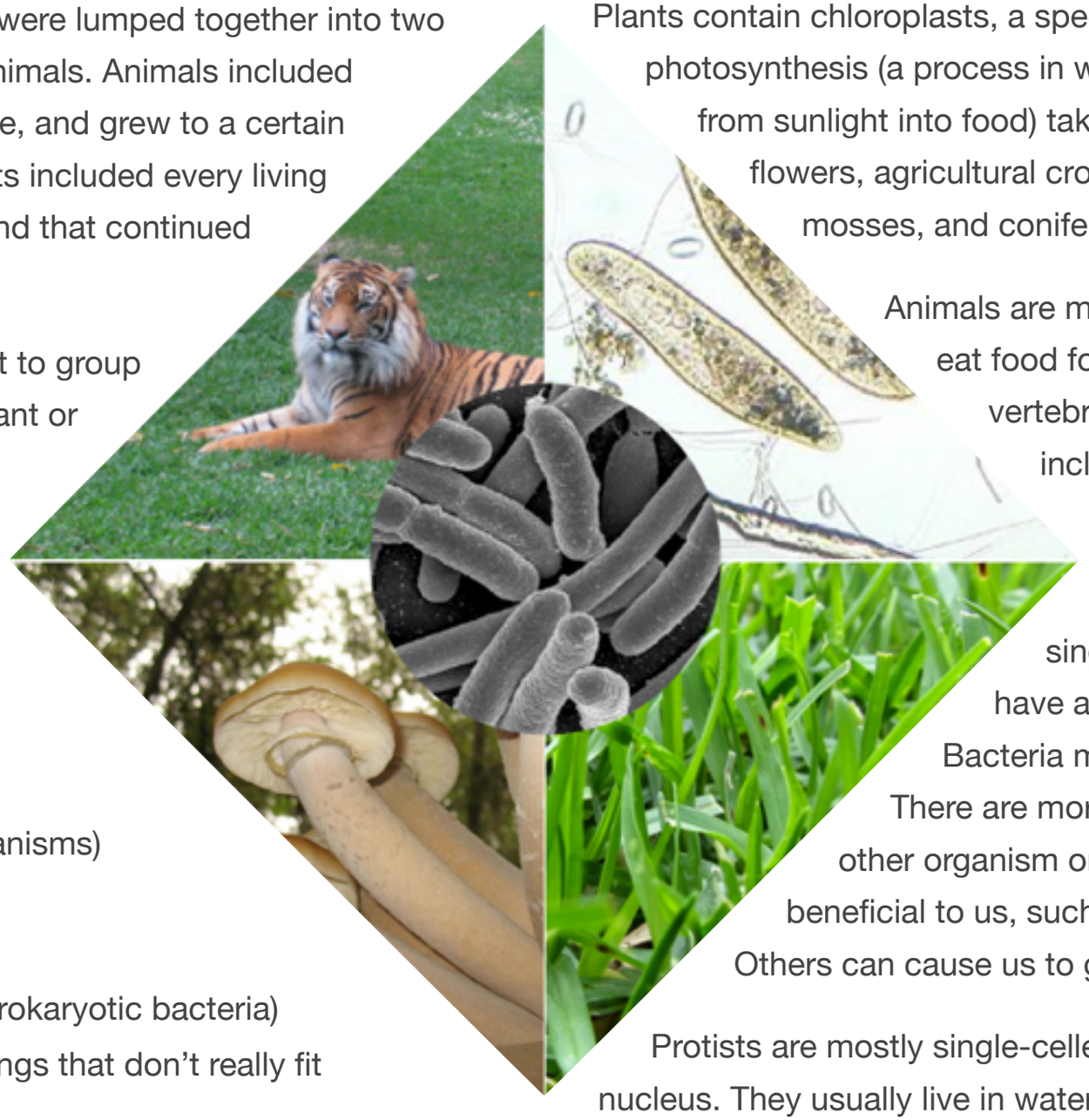
The characteristics of a cell that an organism has enables us to classify that organism into broad groupings called **kingdoms**.

A long time ago all living things were lumped together into two kingdoms, namely plants and animals. Animals included every living thing that moved, ate, and grew to a certain size and stopped growing. Plants included every living thing that did not move or eat and that continued to grow throughout life.

However, it became very difficult to group some living things as either a plant or animal, particularly after the discovery of the microscope! As a result the two kingdoms were expanded into five kingdoms.

The five kingdoms are:

- **Fungi** (fungus and related organisms)
- **Plantae** (the plants)
- **Animalia** (the animals)
- **Prokaryotae** or Monera (the prokaryotic bacteria)
- **Protoctista** or Protista (the things that don't really fit anywhere else).



Fungi are fairly motionless organisms that absorb nutrients for survival. They include mushrooms, moulds, and yeasts.

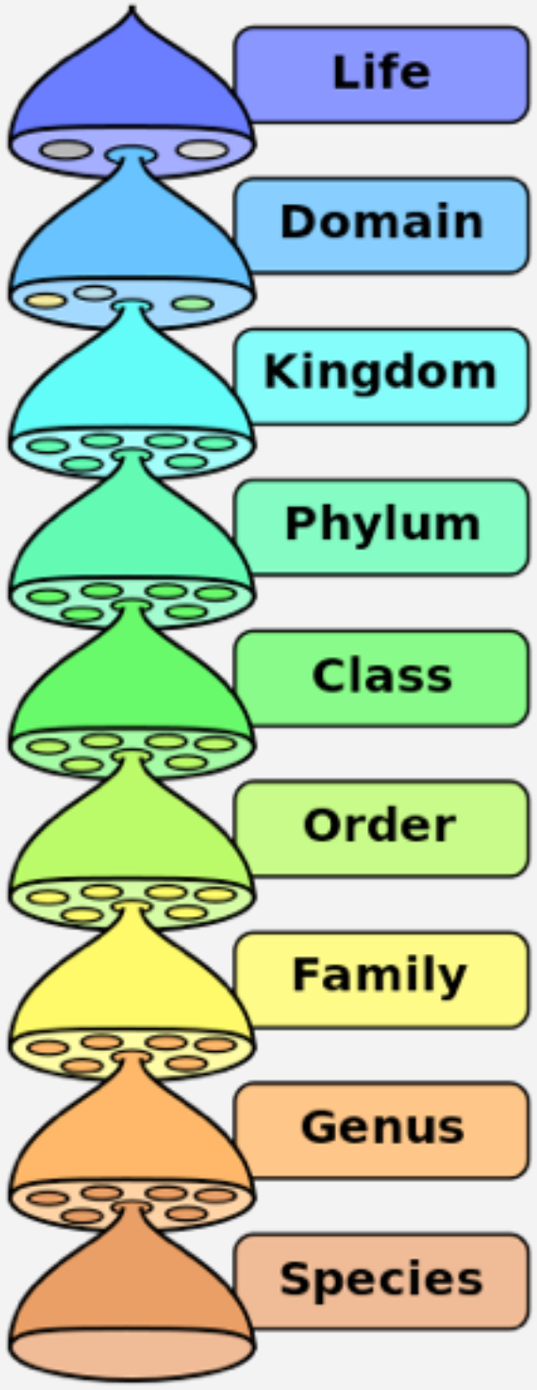
Plants contain chloroplasts, a special organelle where photosynthesis (a process in which plants convert energy from sunlight into food) takes place. They include garden flowers, agricultural crops, grasses, shrubs, ferns, mosses, and conifers.

Animals are multicellular organisms and they eat food for survival. They are divided into vertebrates and invertebrates and include mammals, amphibians, reptiles, birds and fish.

Prokaryotae (Monera) are single-celled organisms that don't have a membrane-bound nucleus. Bacteria make up this entire kingdom. There are more forms of bacteria than any other organism on Earth. Some bacteria are beneficial to us, such as the ones found in yogurt. Others can cause us to get sick.

Protists are mostly single-celled organisms that have a nucleus. They usually live in water. Examples of protists include some algae, paramecium, and amoeba.

Image 1.13 The classification of life



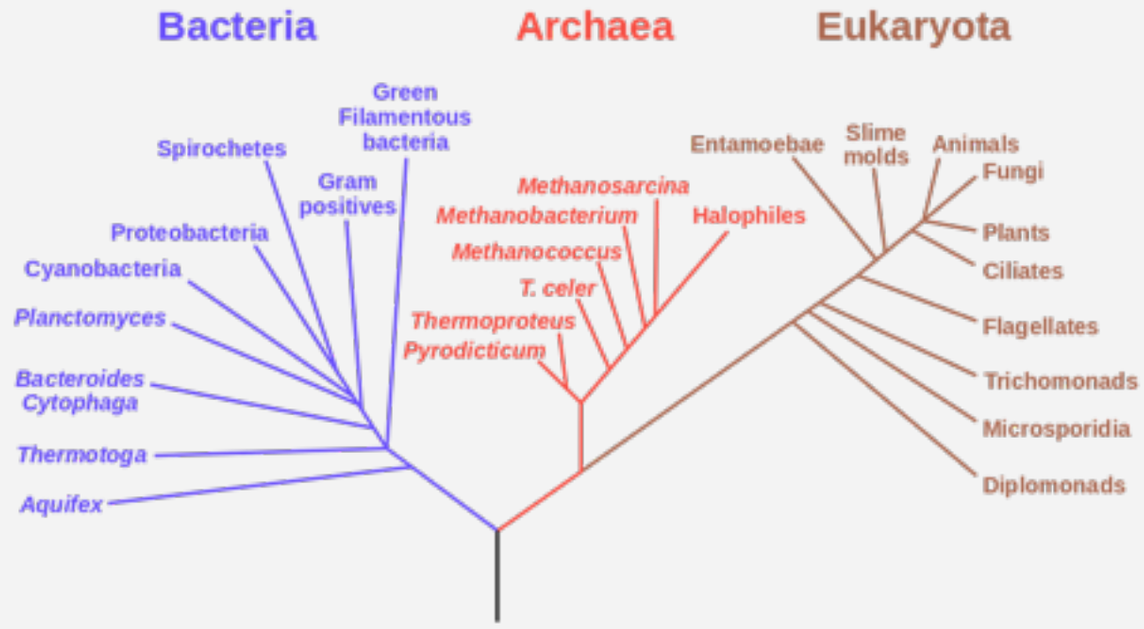
Of course the kingdom an organism falls into forms just one level of classification. All living things can be classified according to their domain, kingdom, phylum, class, order, family, genus and species. For example, humans are classified as: Domain: Eukaryota, Kingdom: Animalia, Phylum: Chordata, Class: Mammalia, Order: Primates, Family: Hominidae, Genus: Homo, Species: H. sapiens.

While we can recognise that there are five kingdoms in which we can classify all organisms, there is actually not complete scientific agreement that there are simply five kingdoms. Some scientists believe that there should be six kingdoms of life while others even think we should classify organisms into eight kingdoms.

One six kingdom method of classification is extremely similar to the five kingdom version. The only difference is that the Prokaryotae (Monera) group has been split into two in the six kingdom system. Thus making the six kingdoms: Animalia, Plantae, Fungi, Protoctista, Archaea, and Bacteria.

Another six kingdom method of classification involves the addition of a new kingdom called Chromista!

Image 1.14 The three domains of life



The three domains illustrate the kingdom Prokaryotae (Monera) split into ‘Bacteria’ and ‘Archaea’.

Table 1.4 Some ways that the cells of different kingdoms vary

	Animalia	Plantae	Fungi	Protoctista	Prokaryotae
Cell Type	Eukaryote	Eukaryote	Eukaryote	Eukaryote	Prokaryote
Number of cells	Multicellular	Multicellular	Multicellular	Mostly unicellular	Unicellular
Chloroplasts	Absent	Present in some cells	Absent	Present in some	Absent
Mitochondria	Present	Present	Present	Usually present	Absent
Nucleus	Present	Present	Present	Present	Absent

EXPERIMENT: EXPLORING CELLS

[To be performed under teacher direction in a science laboratory]

Aim: To observe the differences between a variety of different cell types.

Materials: Various pre-prepared microscope slides with cells from various kingdoms, light microscope.

Method:

1. Pick one of the pre-prepared microscope slides and get it into focus using the microscope.
2. Using pencil, draw a diagram of what you see. Make sure your diagram includes clear and accurate labels (and is large enough to be easy to understand). Diagrams should also have a title with the specimen name and magnification used.
3. Repeat steps 1 and 2 for as many specimens as you can.

Results:

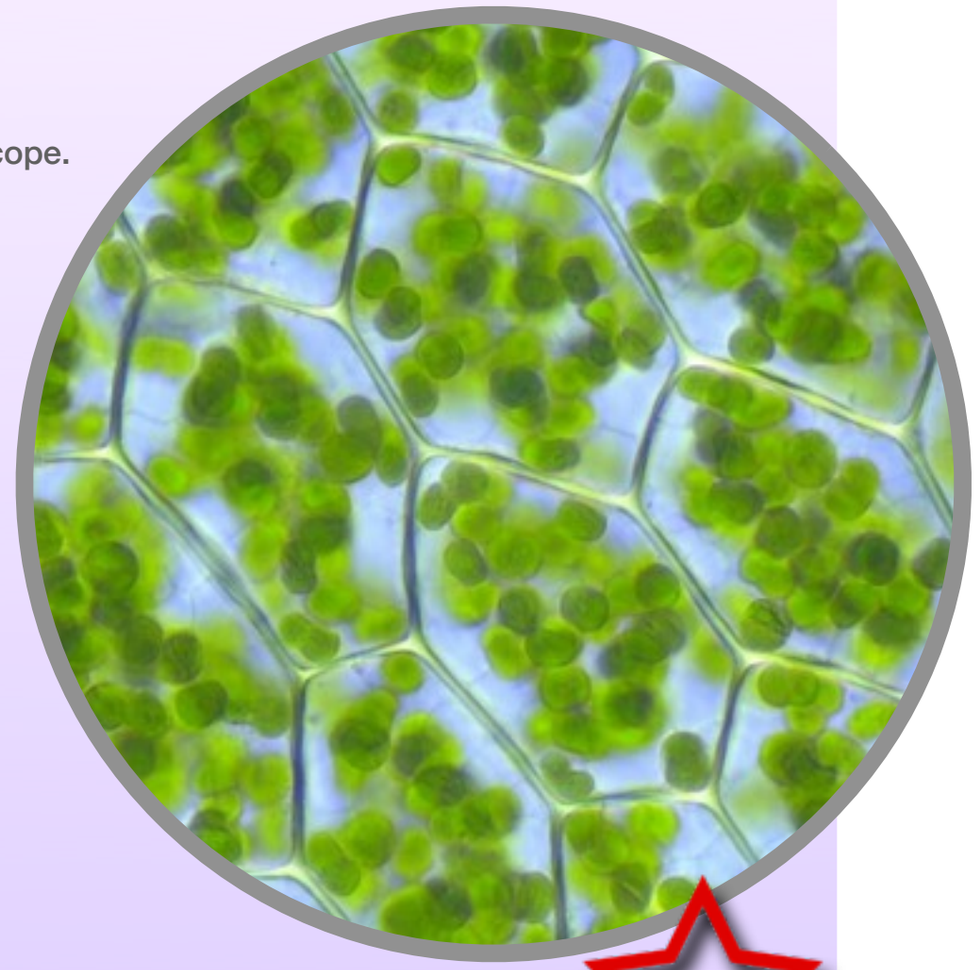
Draw a labelled diagram of each specimen you observed under a microscope.

Discussion:

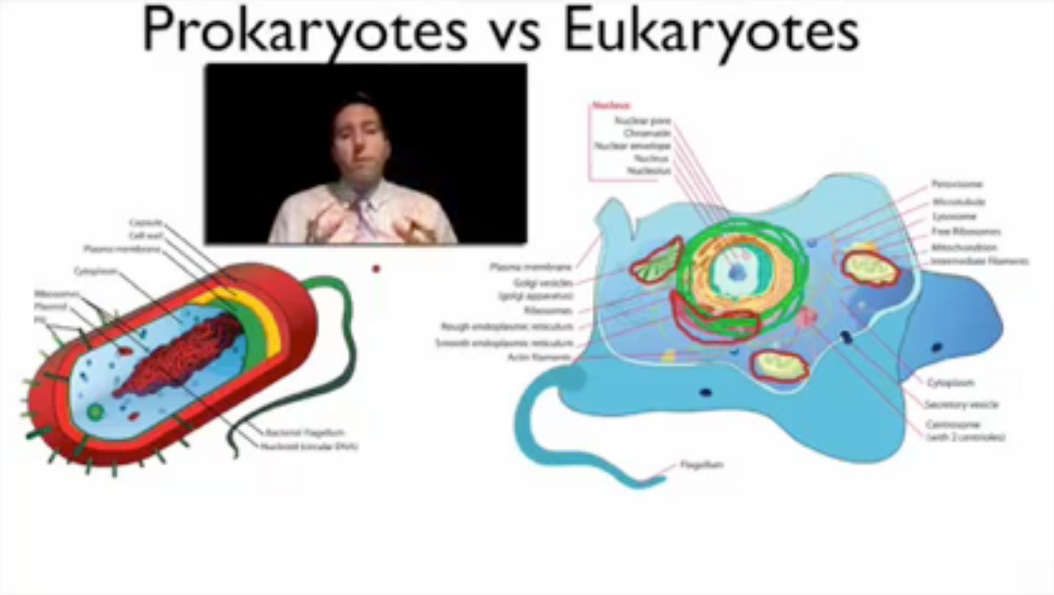
- Explain what you observed during this experiment.
- Describe the feature of the cells you saw and what kingdom of life they come from.
- Discuss anything you could have done better when performing this experiment.

Conclusion:

Sum up your experiment with reference to the aim.



Movie 1.5 How are prokaryotes and eukaryotes different?



Using dot points, summarise the contents of this section.

Movie 1.6 How is life classified?



PROJECT - Make a Cell Model

Create a three dimensional model of a cell or even several different types of cells. Your model should include:

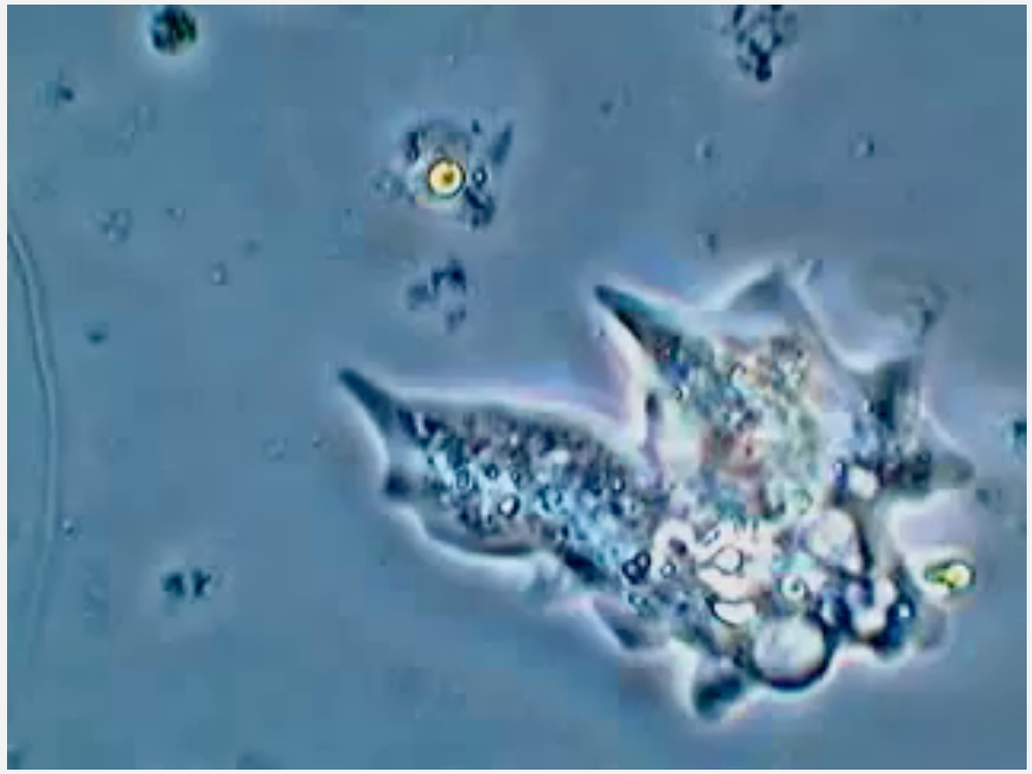
- All of the different parts of the cell.
- All parts should be clearly labelled.
- You should include a key with a brief description of the function of each organelle/cell part.
- A clear heading indicating what your model is of.
- Your model should be at least A4 in size.

6



(You may even like to make a cell cake using icing for the cytosol and lollies for organelles.)

Movie 1.7 A single celled amoeba



This clip shows a single celled amoeba, from the kingdom Protoctista, with many organelles visible inside it.

QUESTIONS

1. Create a venn diagram comparing prokaryotic and eukaryotic cells.
2. Draw a simple labelled diagram of a cell including only the common parts that all cells share, that is, a cell membrane, cytoplasm, ribosomes, and DNA.
3. When you consider the three domains of life, do you think it is more appropriate to have five or six kingdoms of life. Justify your answer.
4. Find out the kingdom, phylum, class, order, family, genus and species for three different species.
5. Name an organism from each of the five kingdoms using the correct scientific name.
6. In your own words, describe how the ocular and objective lenses combine to form the total magnification of a microscope.

Section 3

Do different parts of cells do different jobs?

We can think of the cell as a microscopic city. Like a real city it requires many services to keep it clean and running smoothly. Think of some of the services a real city needs: energy provision, waste disposal, and a delivery service just to name a few.

If we look inside a complex eukaryotic cell we see that that there are many different types of **organelles**. We can think of organelles in a cell as being similar to organs in our body. Just as our organs have specific functions. For example, our heart pumps blood or our kidneys get rid of wastes. The organelles of a cell have specific functions too.

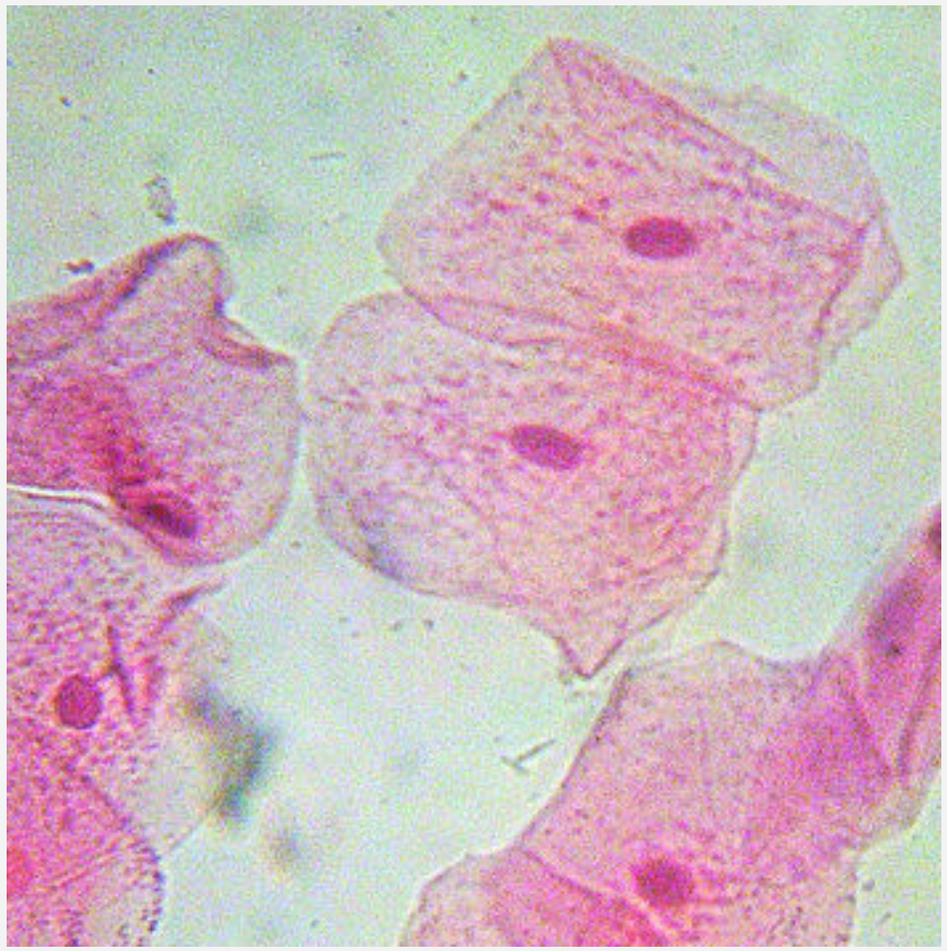
Or, like our imagined city, we can think of organelles as being the "workers" that provide the services a eukaryotic cell needs.

The following is a list describing the various functions of some common organelles.

Nucleus

The **nucleus** can be thought of as the chief of the city or the city mayor. It directs all cell activities and is the control centre of the cell. The nucleus is where the genetic material, or **DNA**, is kept. This DNA provides the instructions for all of the cells activities.

Image 1.15 Human cheek cells

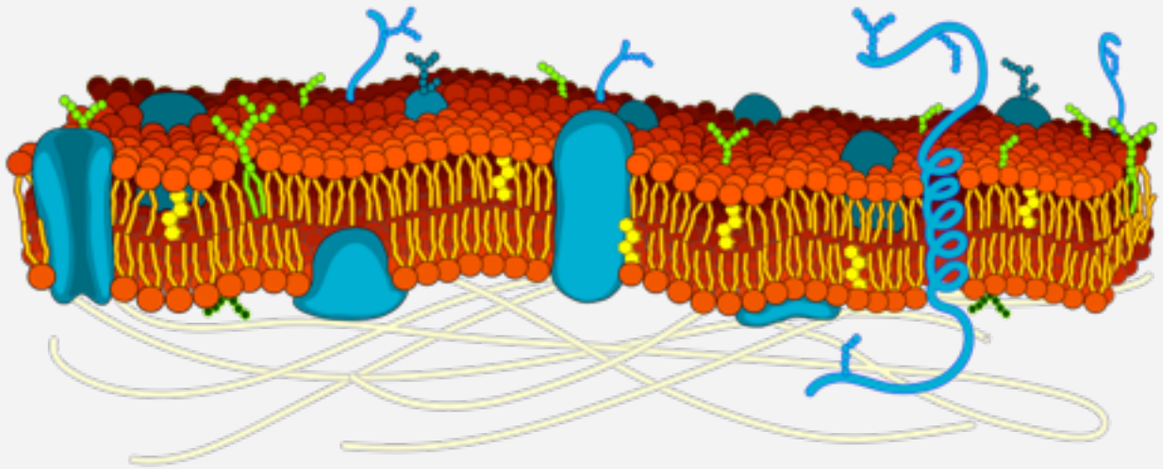


The nucleus of these stained cells is clearly visible as a dark pink oval in the middle of each cell.

Cell membrane

Like a city perimeter, the **cell membrane** surrounds the cell and keeps everything inside it. Cell membranes also have the ability to regulate the entrance and exit of substances, thereby controlling what is inside the cell. These membranes also protect the cell from outside forces.

Image 1.16 A detailed diagram of a cell membrane

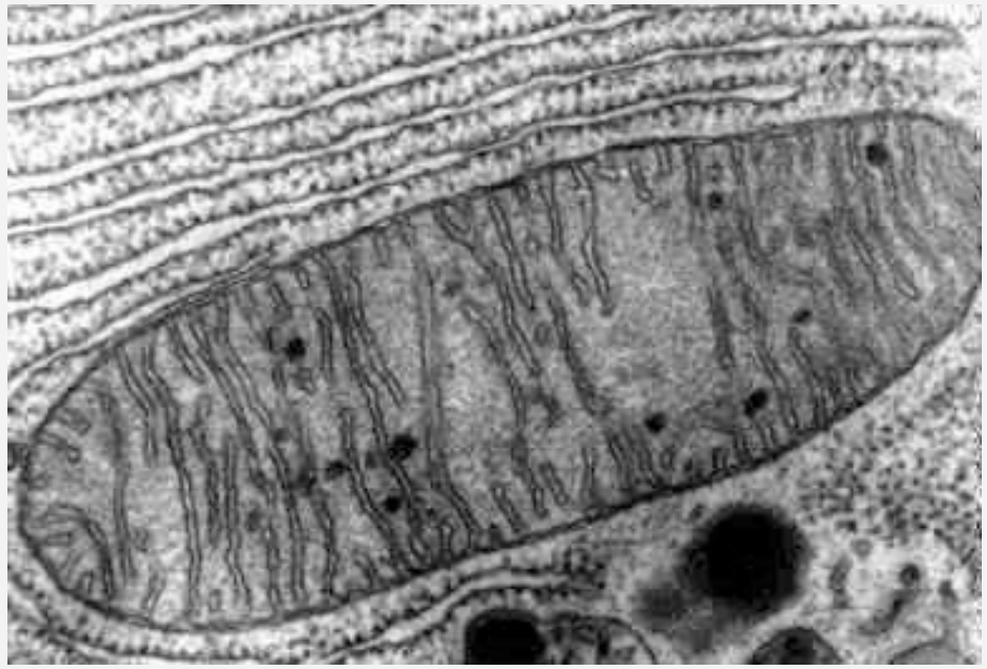


When you think about a cell membrane you can imagine it is like a big plastic bag with some tiny holes. That bag holds all of the cell pieces and fluids inside the cell and keeps any nasty things outside the cell. The holes are there to let some things move in and out of the cell.

Mitochondrion

The **mitochondria** are the main energy source of the cell, in fact, they are often called the "power plants" of the body because this is where energy for the cell is generated.

Image 1.17 A mitochondrion



As mitochondria are so small, this image was captured using an electron microscope.

The process of creating cell energy is known as **cellular respiration**. Most of the chemical reactions involved in cellular respiration happen in the mitochondria.

Chloroplast

Chloroplasts are the food producers of the cell. Importantly, they are only found in plant cells and some protists.

Like a solar energy powered factory, they use sunlight to create food for the city. Chloroplasts are the site of photosynthesis a process in which the plant uses carbon dioxide, water and sunlight to create food in the form of sugar for the plant cell.

Movie 1.8 Chloroplasts



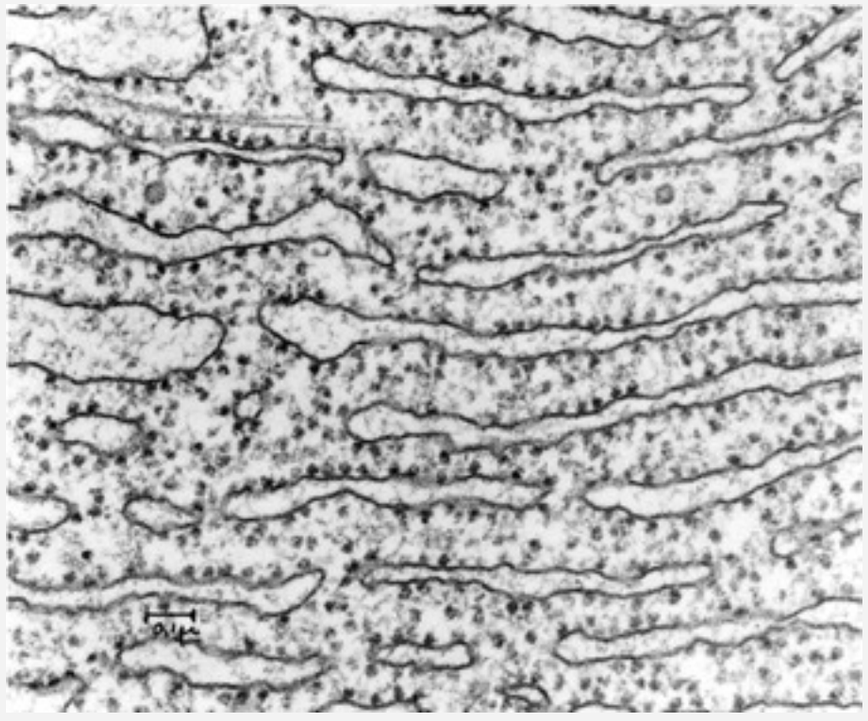
Elodea is a species of water plant. Its cells contain lots of green chloroplasts which can be seen here moving about the cell via a process called cytoplasmic streaming.

Ribosome

Ribosomes are very small organelles and they are responsible for protein synthesis. Protein synthesis is very important to cells, therefore large numbers of ribosomes are found in cells. Ribosomes float freely in the cytoplasm, and are also bound to the endoplasmic reticulum (ER). Endoplasmic reticulum bound to ribosomes is called rough ER because the ribosomes on the endoplasmic reticulum give it a rough sandpaper like look.

Ribosomes could be thought of as the builders of our city. They bring together all the raw ingredients such as RNA (copies of the original DNA blueprints) and amino acids to assemble proteins. The proteins created are essential to cell and organism function. Think of proteins as machinery for cell functions much like electricity and plumbing are essential in a real city. For example, enzymes are a type of protein without which life could not exist.

Image 1.18 Ribosomes



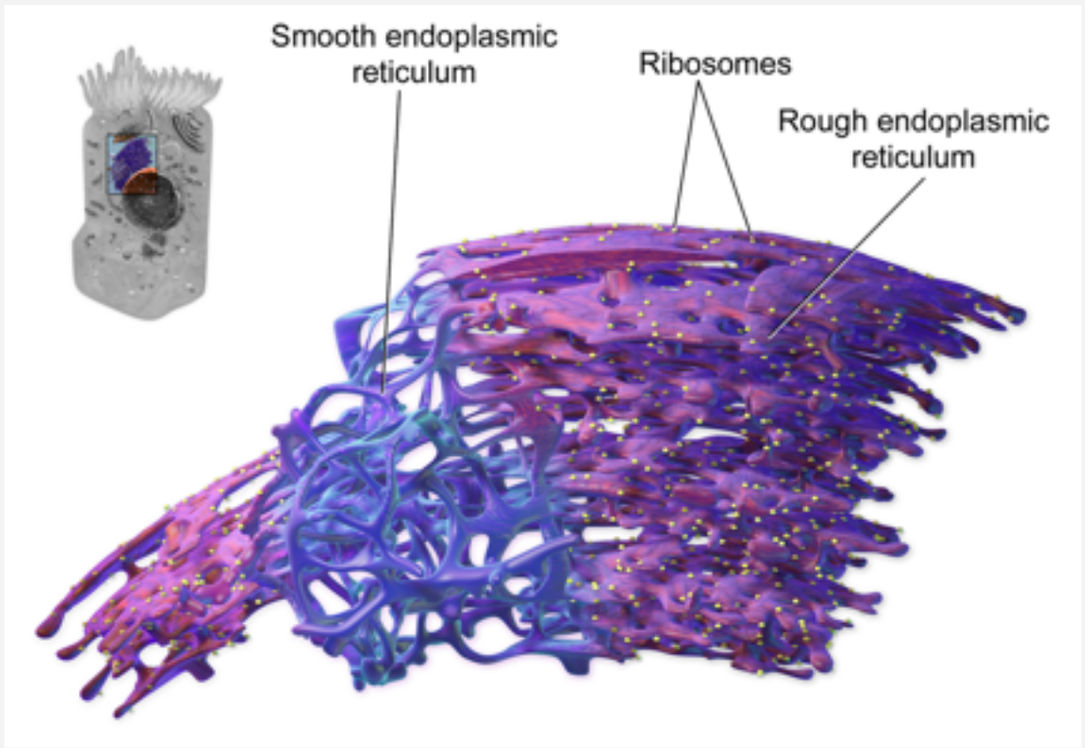
An electron microscope image showing endoplasmic reticulum decorated with ribosomes.

Endoplasmic Reticulum

The **endoplasmic reticulum** is, in essence, the transportation system of the eukaryotic cell. Proteins are manufactured and shipped to needed areas via this transportation system.

There are two types of endoplasmic reticulum, smooth and rough.

Image 1.19 The endoplasmic reticulum



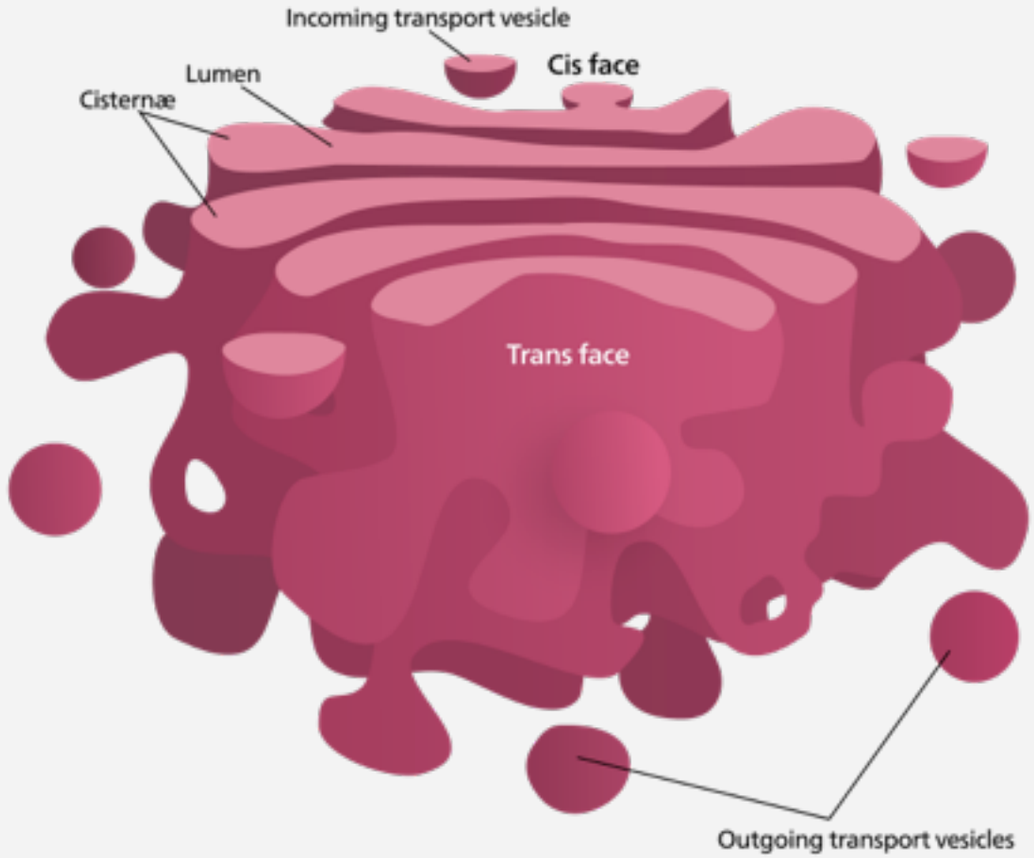
The surface of the rough endoplasmic reticulum is studded with protein-manufacturing ribosomes giving it a "rough" appearance (hence its name).

The smooth endoplasmic reticulum lacks ribosomes.

Golgi Apparatus

Like a post office, the **golgi apparatus** is used for shipping those goods created by the endoplasmic reticulum and ribosomes to the rest of cell. They ship all of the materials in tiny transport vesicles like little membrane bound bubbles.

Image 1.20 Golgi Apparatus



Vacuole

Vacuoles are just membrane sacs. You can think of them as containment units or warehouses for anything in excess in our city. They can hold many substances from organic molecules to simple excess water. Plant cells have a large central vacuole that is important in maintaining plant cell shape.

Image 1.21 Plant cell vacuole



This plant cell diagram shows a clearly visible large central vacuole.

Lysosomes

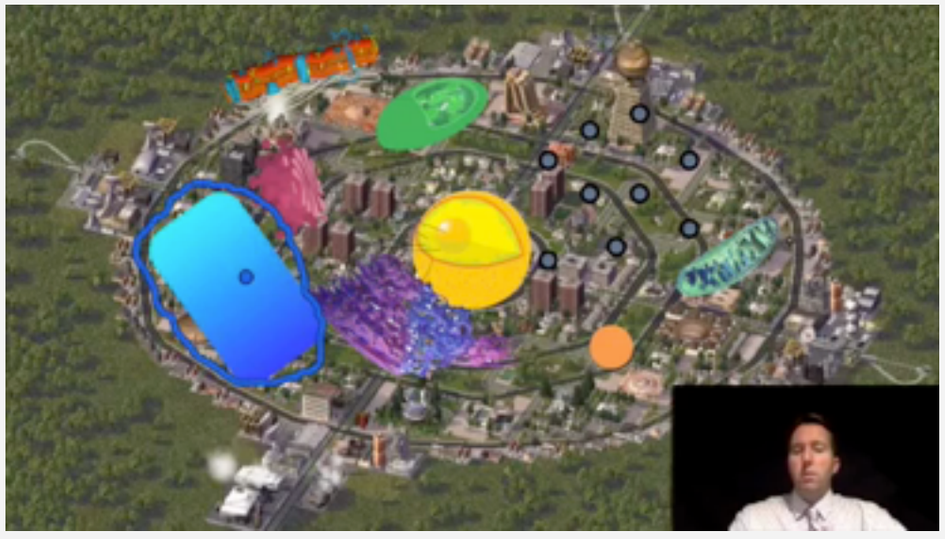
The **lysosome** is the waste disposal or digestive system in the cell. It breaks down molecules using special digestive enzymes. Like waste disposal in a city, lysosomes help keep excessive or bulky macromolecules from building up in the cell.

This demonstrates why it is important for all parts of a cell to be compartmentalised. In this case the cell couldn't use the destructive enzymes if they were not sealed off from the rest of the cell.

Cytoplasm

The **cytoplasm** is all of the material found inside a cell excluding the nucleus. So the cytoplasm comprises the organelles as well as a jelly-like substance called the **cytosol**. The cytosol is like the cell landscape, found in any space where organelles are not and therefore is much like the lawns and parks of our city.

Movie 1.9 Organelles explained



Preparing a wet mount

To view many specimens under a light microscope it is first necessary to prepare a microscope slide for viewing. This is called preparing a wet mount.

To make a wet mount:

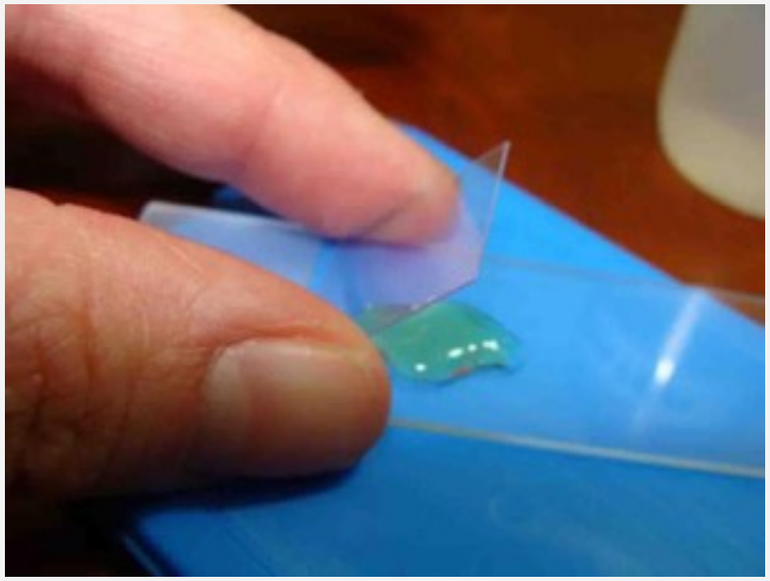
1. Place your sample on the slide. Note it should be extremely thin to be viewed under a light microscope.

2. Using an eyedropper or pipette, put a drop of water on the sample. If you are dying your sample you can place a drop of the dye on top of it.

3. Place one end of the cover slip on the slide and gently drop the coverslip onto the sample. You can also lower the coverslip onto the sample using the end of a toothpick. This will help to prevent air bubbles from getting trapped under the cover slip.

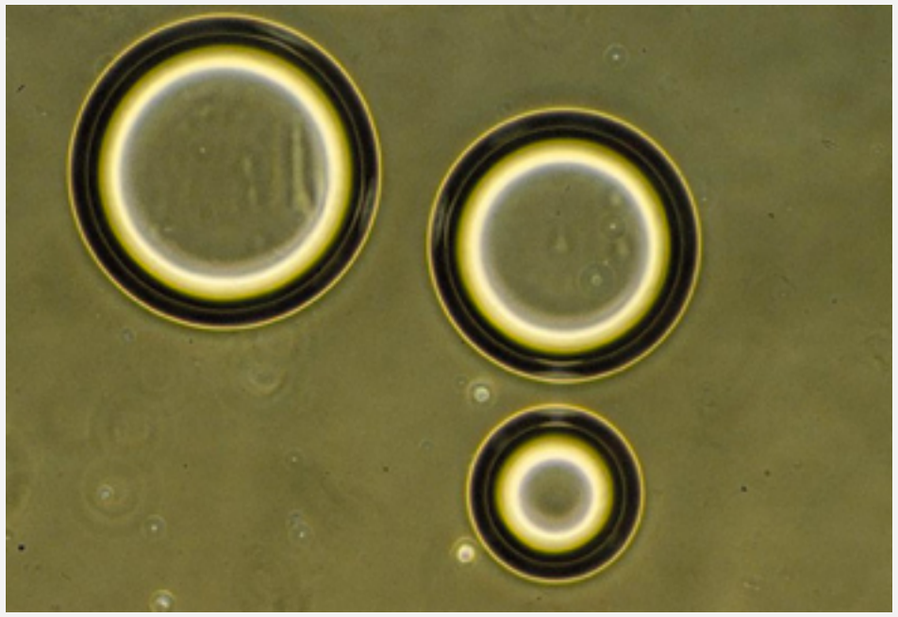
4. The water (or dye) should just fill the space between the cover slip and the slide. If there is too much liquid and the cover slip is floating around, remove some water by holding the edge of a paper towel next to the edge of the cover slip. If there is too little liquid and some of the space under the cover slip is still dry, add more liquid by placing a drop right next to the cover slip. A little practice will help you learn how much liquid to add.

Image 1.22 Preparing a wet mount

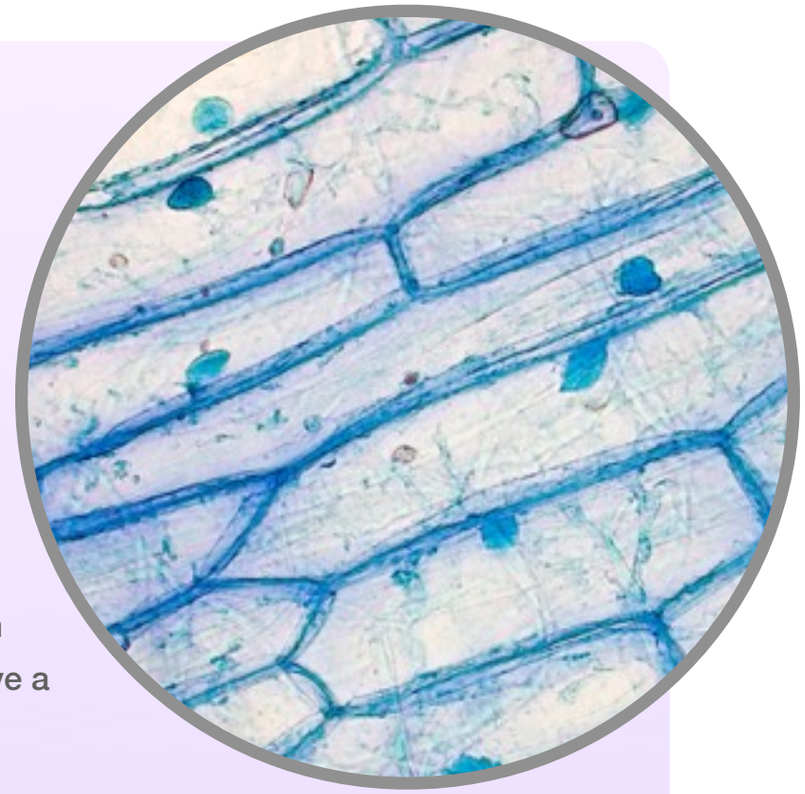


Th coverslip should be gently lowered onto the sample at an angle to reduce the chance of bubbles forming.

Image 1.23 Air bubbles under the microscope



Sometimes air bubbles can become trapped under the coverslip. Do not confuse these as cells or organelles.



EXTENDED EXPERIMENT: EXPLORING ORGANELLES

[To be performed under teacher direction in a science laboratory]

Aim: To observe the various organelles found inside different cells and learn more about their specific function.

Materials: Light microscope, pre-prepared microscope slides with cells stained to see one or more organelles, elodea leaf, banana, iodine solution, onion, tweezers, methylene blue, live amoeba or paramecium, microscope slides, cover slips, plastic pipette or eyedropper, paper towel/tissue, water.

Method:

1. For all specimens you observe, and using pencil, draw a diagram of what you see. Make sure your diagram includes clear and accurate labels (and is large enough to be easily understood). Diagrams should also have a title with the specimen name and magnification used.
2. Pick one of the pre-prepared microscope slides and get it into focus using the microscope.
3. Repeat step 2 for any other pre-prepared slides you have access to.
4. Peel off an extremely thin 'see-through' section of onion and prepare a wet mount using methylene blue to stain the nucleus a dark blue colour.
5. Squash a small amount of banana onto a slide and prepare a wet mount using iodine to stain the sugar in the vacuoles a dark purple colour.
6. Prepare a wet mount of an elodea leaf to view its chloroplasts.
7. Prepare a wet mount of a live amoeba or paramecium, try and discern any organelles.



Results:

Draw a detailed, labelled diagram of each specimen you observed under a microscope.

Discussion:

- Explain what you observed during this experiment.
- Describe the features and functions of the organelles you saw.
- Explain why you think some cells have different types and different numbers of organelles. For example, why do banana cells have many sugar filled vacuoles and no chloroplasts while elodea leaf cells have many chloroplasts?
- Discuss any difficulties you had and anything you could have done better when performing this experiment.

Conclusion:

Sum up your experiment with reference to the aim.

Song 1.1 The Cell Song

Listen close to the story I tell. It's the rapping story of the living cell. It's a happy tune that's sort of cheery. About a real tough topic called the cell theory.

All animals, plants, and protists too, Are made of cells with different jobs to do. They're the basic units of all organisms, And I hope by now you got the rhythm.

It all started with one dude named Hooke. Who at some cork cells took a look. He used a scope and took his time. 'Cause a cell is small and thinner than a dime.

Say 1, 2, 3, 4, Are you ready to learn some more? The animal cell has many parts, And you must know each one by heart.

Like the farmer man in the dell. The nucleus controls the cell. its gives the orders -- kind of like a brain. And it's protected by a nuclear membrane.

Around the cell, you'll find another "skin," The plasma membrane holds the whole cell in, But its job isn't simple there's no doubt, It lets some particles go in and out.

Now please don't lose your science enthusiasm, Listen to the story of the cytoplasm. All around the cell this fluid does go, But in the nucleus it will not flow.

And don't forget a ribosome, This is where proteins come from. These protein factories are small, you'll agree, So small you'll need an electron microscope to see.

Just when you thought you weren't having any fun, Along comes the endoplasmic reticulum. These tubelike structures serve as a track, To carry stuff to the membrane and back.

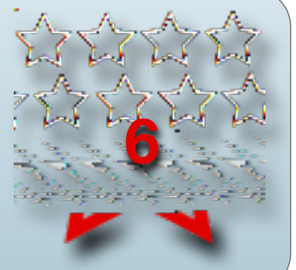
Now have you ever seen any doughnuts without holes? In a cell, they're called vacuoles. They're filled with stuff like H_2O, And they carry food so the cell can grow.

Last of all, but not the very least, Mitochondria - mighty cellular beasts, Since they turn sugars into energy so well, We call them the powerhouse of the cell.

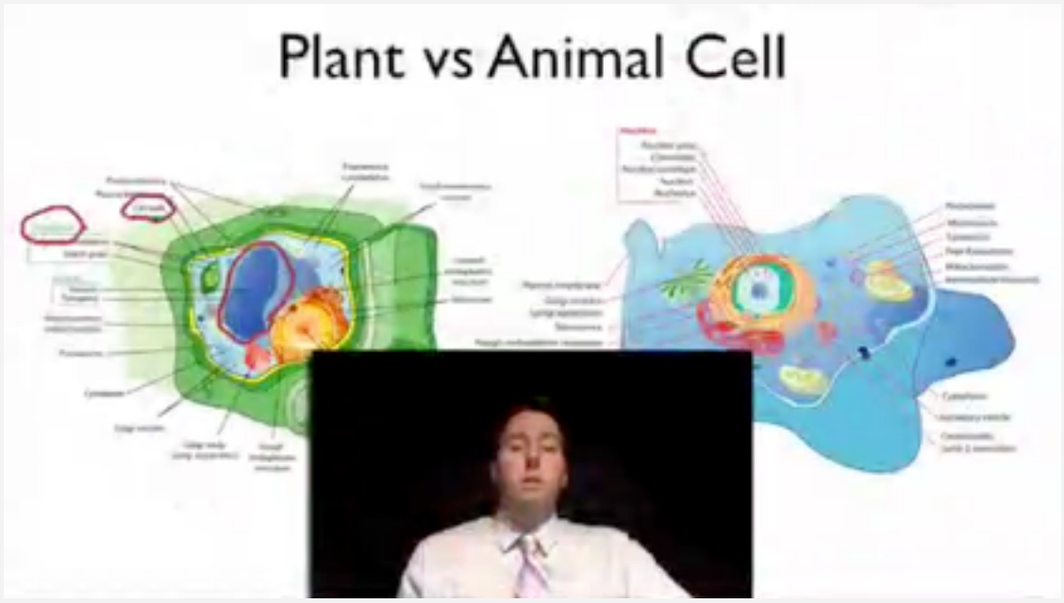
Now my friend, you know it well, The unforgettable story of the living cell.

ACTIVITY

Create your own song about cells.



Movie 1.10 How are plant and animal cells different?



Using dot points, summarise the contents of this section.

QUESTIONS

1. Think about all of the different parts of a cell and decide which part is the most important. Justify your answer.
2. Find out what the 'endosymbiotic theory' is and how it explains how organelles originally became a part of eukaryotic cells.
3. Create a table listing at least four organelles and their functions.
4. Find out the chemical equation for the process that happens in the mitochondria (aerobic cellular respiration) and the chemical equation for the process that happens in the chloroplast (photosynthesis).

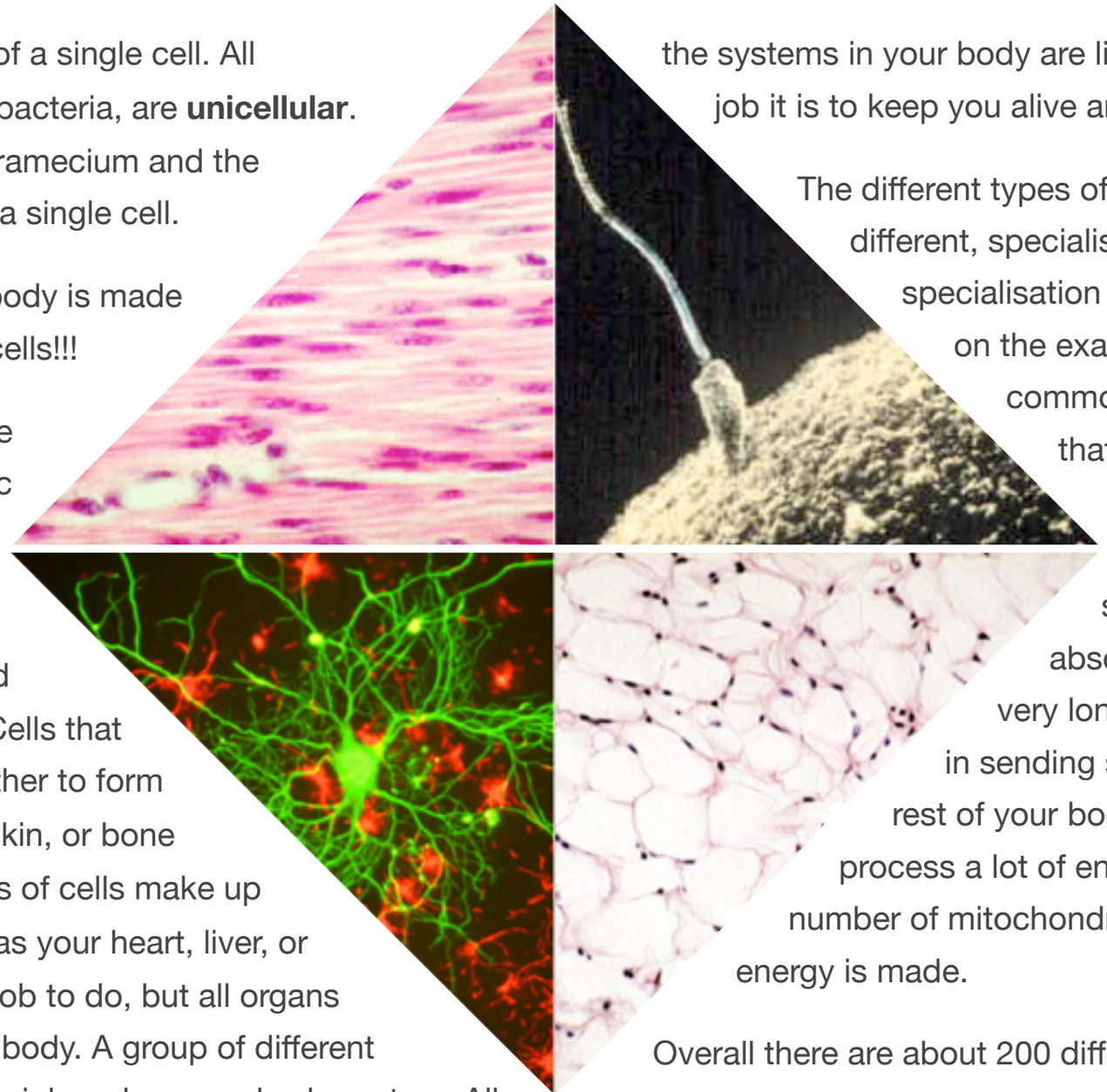
How do human cells vary?

Some organisms are made up of a single cell. All **prokaryotes**, and therefore all bacteria, are **unicellular**. Similarly, organisms like the paramecium and the amoeba are composed of only a single cell.

On the other hand the human body is made up of approximately 40 trillion cells!!!

Over millions of years cells have evolved to perform very specific roles in **multicellular** organisms like a human.

Each type of **cell** has a size and shape that is suited to its job. Cells that do the same job combine together to form body **tissue**, such as muscle, skin, or bone tissue. Groups of different types of cells make up the **organs** in your body, such as your heart, liver, or lungs. Each organ has its own job to do, but all organs work together to maintain your body. A group of different organs working together to do a job makes up a body system. All the systems in your body are like members of a team whose job it is to keep you alive and healthy.



The different types of cells in your body have different, specialised jobs to do. The specialisation of cells depends almost always on the exaggeration of properties common to cells. For example, cells that line the intestine have extended cell membranes. This increases the amount of surface area that is available to absorb food. Nerve cells can be very long, which makes them efficient in sending signals from the brain to the rest of your body. Cells in heart muscle process a lot of energy, so they have a large number of mitochondrion, the part of the cells where energy is made.

Overall there are about 200 different kinds of specialised cells in the human body. Some examples of specialised cells include:

Bone cells

Bone cells are found within the bone tissue and are responsible for the make-up of our skeleton. Bone cells do many things for the skeletal system such as the development of new bones, the maintenance of bones, and the regulation of minerals in the body.

Image 1.24 Bone cells

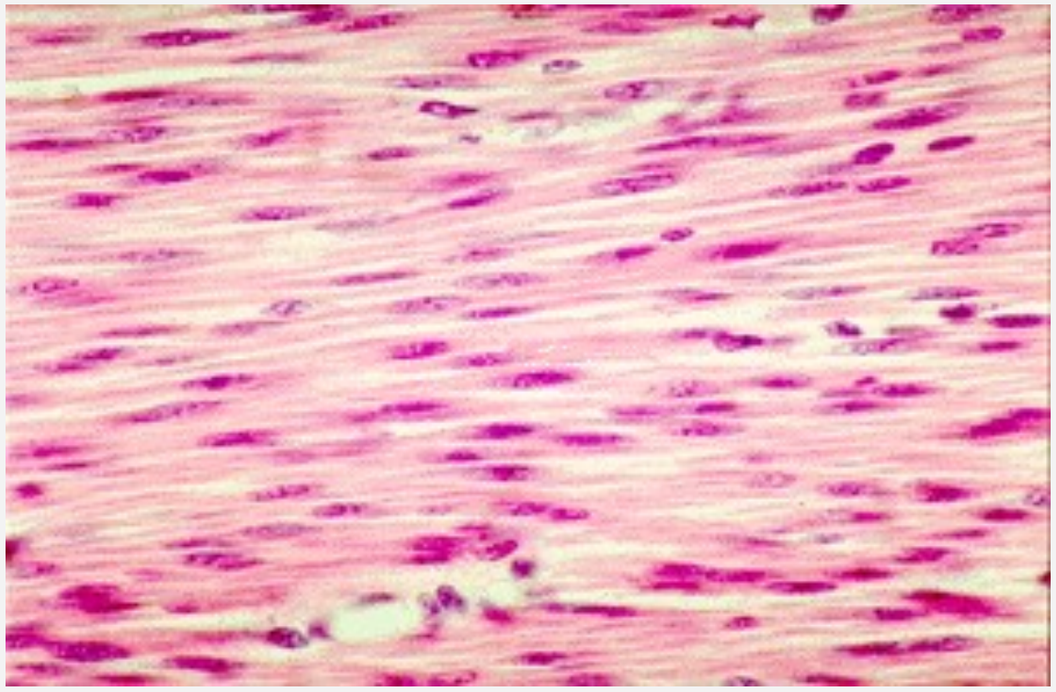


Osteoblasts build bones and osteocytes hold the bone together.

Muscle cells

Muscle cells are mostly long and have the ability to contract and relax. When muscle cells and therefore the muscle tissue contracts, the muscle shortens and causes movement. Often, when one muscle contracts another one relaxes and is lengthened. Muscle cells have many mitochondria, which allows them to respond quickly to the need for doing work.

Image 1.25 Muscle cells (or myocytes)

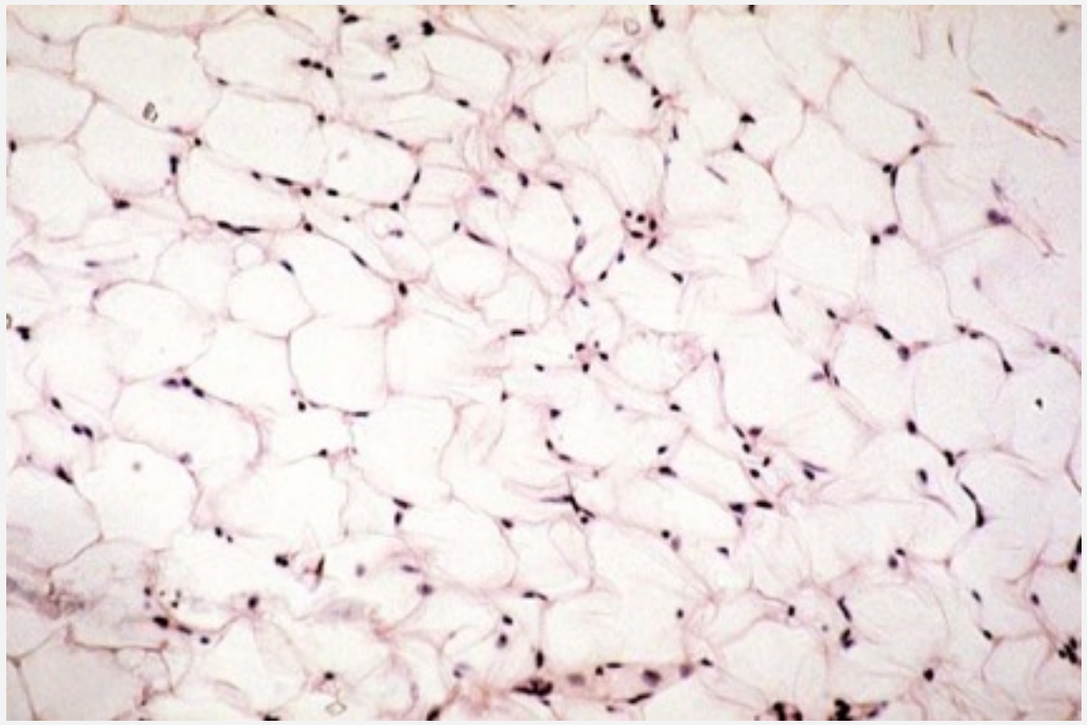


Fat cells

Fat cells, or adipocytes, are basically like a tiny plastic bag that holds a drop of fat. Fat cells are large cells that have very little cytoplasm (only 15 percent of the cell volume) a small nucleus

and one large fat droplet that makes up about 85 percent of the cell volume.

Image 1.26 Fat cells

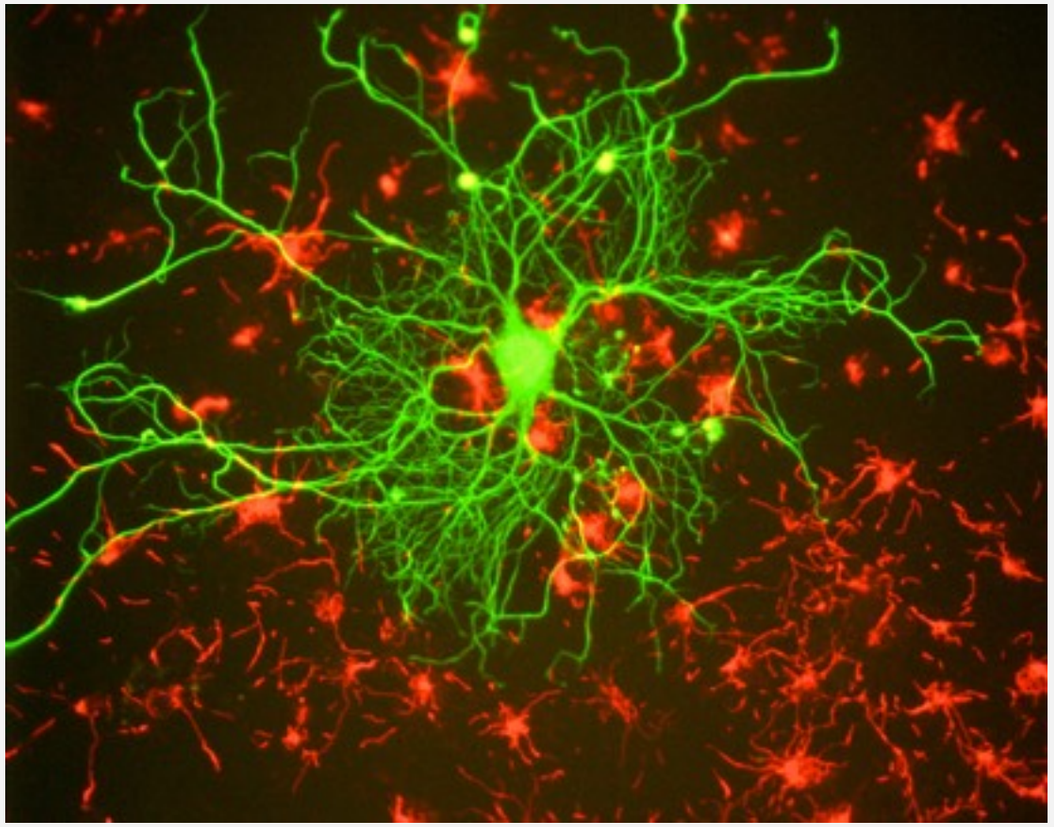


In these fat cells the darkly stained nucleus can be seen but the majority of the volume of the cell is made up of a large white fat droplet.

Nerve cells

Nerve cells, or neurons, are specialised to carry messages throughout the body. The human brain has approximately 100 billion neurons. Nerve cells can be microscopic in size or up to a metre long in humans. Nerve cells carry messages (such as sight, touch or smell) from the external environment to the brain. The

Image 1.27 A nerve cell (neuron) shown in green



brain also sends messages to muscles using nerve cells. Nerve cells have many extended projections to both receive and deliver messages.

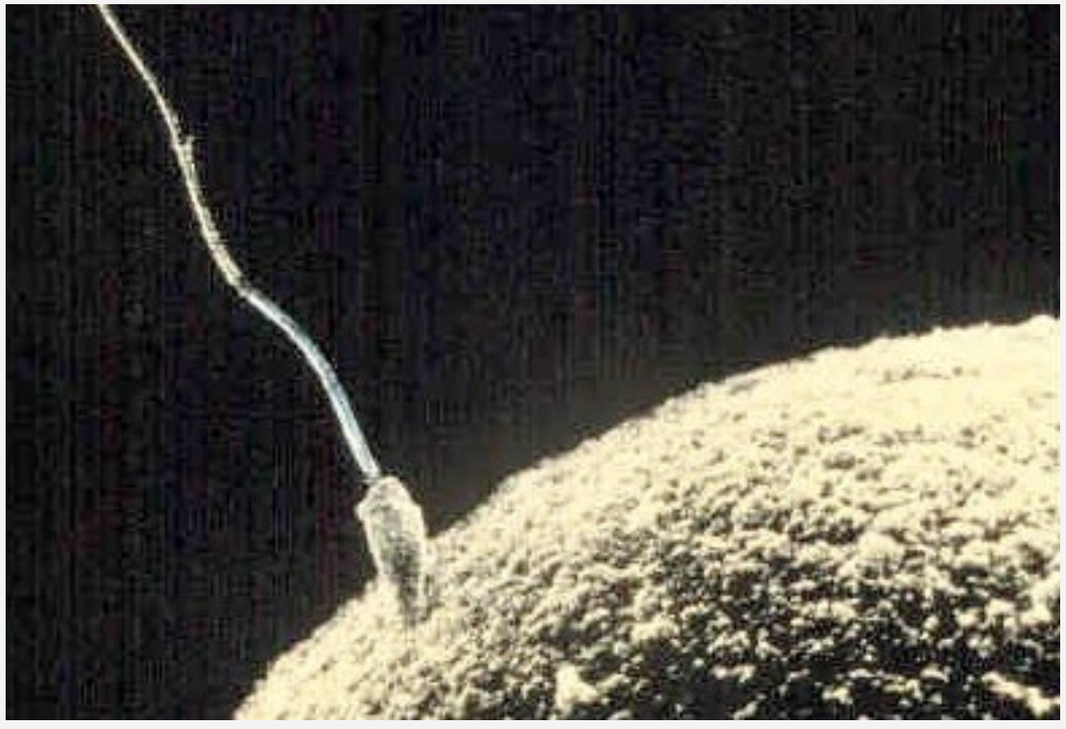
Secretory cells

Secretory cells release chemicals from themselves. Some types are found in the digestive system and they secrete digestive enzymes and gastric acid. Salivary cells in the mouth release saliva. Hormone secreting cells secrete their products directly into the bloodstream.

Reproductive cells

A **reproductive cell**, or gamete, is a cell that joins with another cell during sexual reproduction. A male gamete called a sperm fuses with a female gamete called an egg to form the beginning of a new organism.

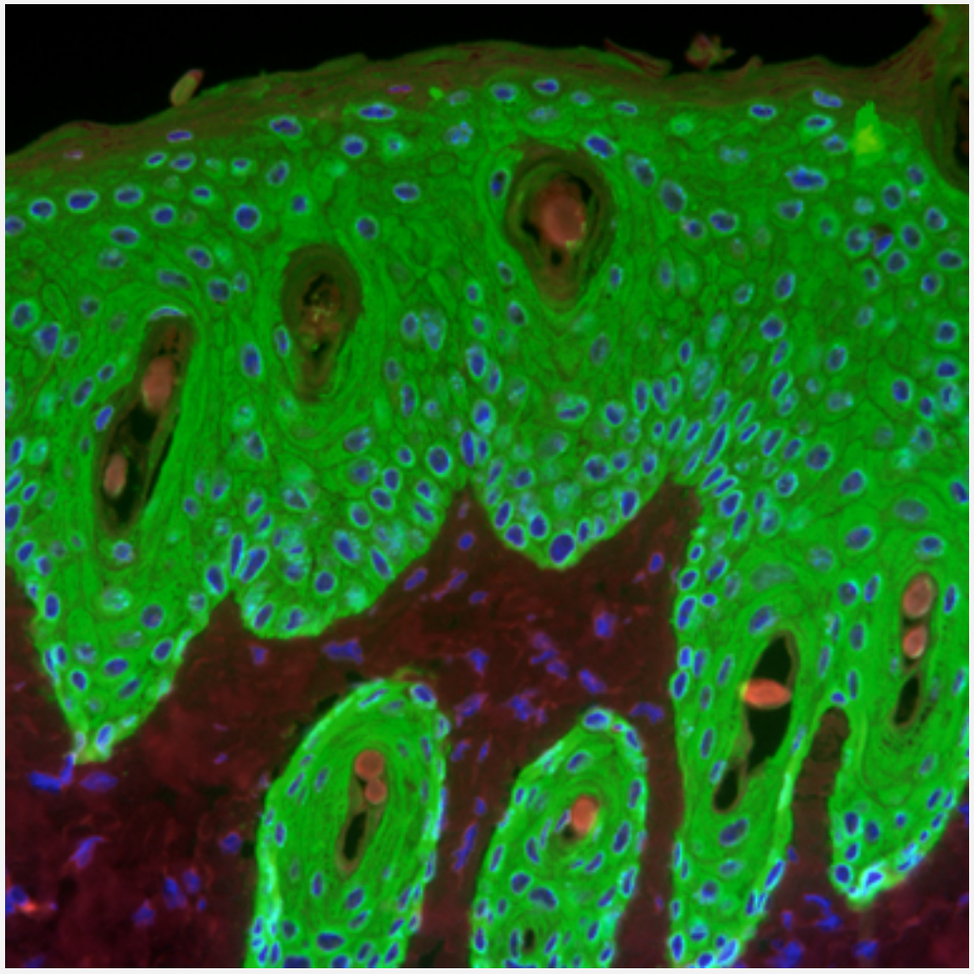
Image 1.28 A sperm cell meets an egg cell



Skin cells

Skin cells make up the largest organ of our body, the skin. The primary function of the outer skin cells (keratinocytes) is the formation of a barrier against environmental damage. This damage can come from such things as pathogens (bacteria,

Image 1.29 Skin cells



Skin cells (keratinocytes) dyed green from the skin of a mouse.

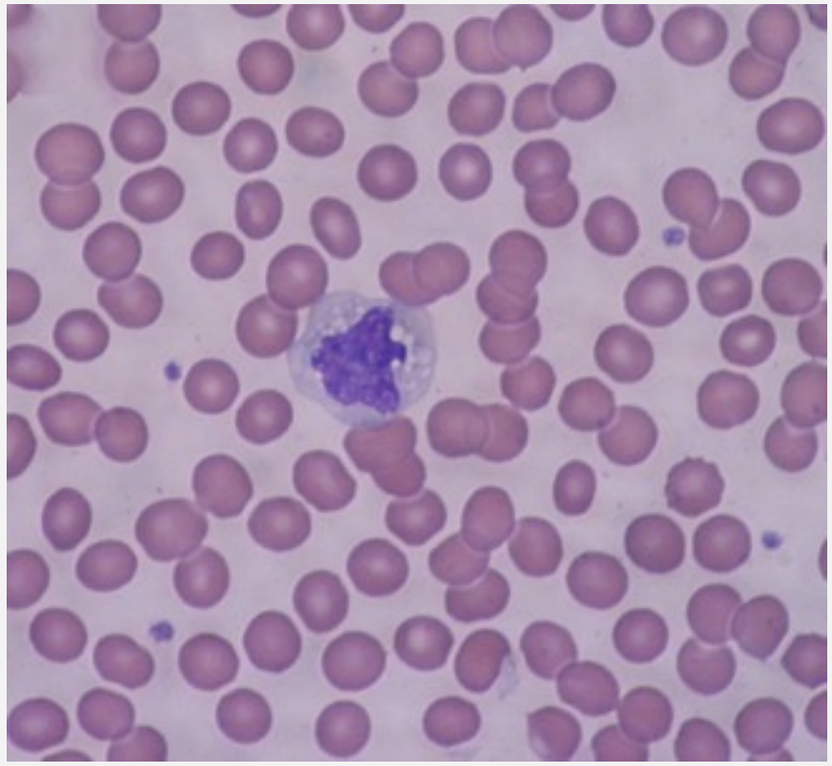
fungi, parasites, viruses), heat, UV radiation and water loss. Skin cells are also involved during the process of wound healing.

Blood cells

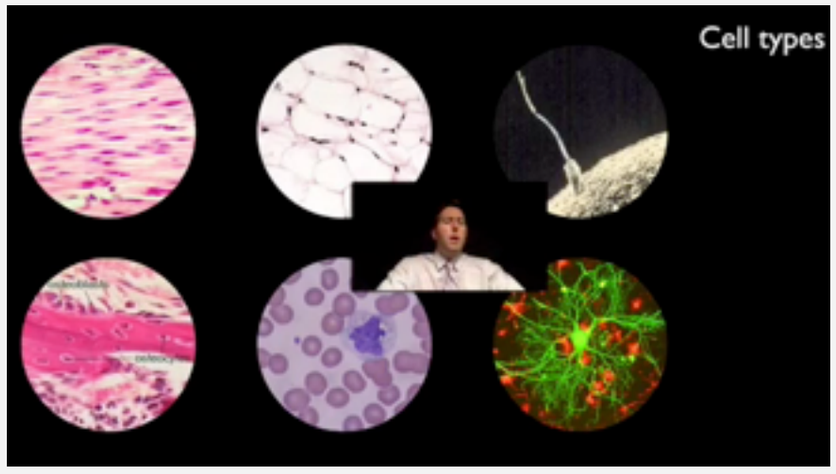
There are several types of **blood cells**. Red blood cells carry oxygen around the body, white blood cells fight disease and platelets clot the blood if you get cut. Red blood cells are

indented in the middle to increase their surface area and therefore increase their capacity to collect oxygen from the lungs.

Image 1.30 A single white blood cell (dark purple) surrounded by red blood cells



Movie 1.11 Are there different types of cells?



QUESTIONS

1. Explain why it is beneficial for different cells in a multicellular organism to have different functions.
2. If you were a scientist from the future capable of designing any type of specialised cell with any function, what is one type of cell you would design to improve human capabilities. Explain why this new cell would be of benefit and if this cell type would form part of an existing organ or form a new type of organ or body system.
3. Re-write the following words in order from the smallest thing to the largest thing:
   Body, cell, atom, organ, tissue, population.
4. Research the names of three different types of secretory cells.
5. If you had to choose to lose one type of cell from your body for ever, which type would it be and why. Choose from:
   Bone cells, Muscle cells, Fat cells, Nerve cells, Secretory cells, Reproductive cells, Skin cells, or Blood cells. Remember to explain your choice.

Using dot points, summarise the contents of this section.

PROJECT - Exploring Stem Cells

We know that the human body is made up of many different kinds of specialised cells such as muscle cells, nerve cells, fat cells and skin cells. But how do these different cell types form?

Scientists have discovered that all specialised cells originate from stem cells. A stem cell is a cell that is not yet specialised. The process of specialisation is called differentiation and once the differentiation pathway of a stem cell has been decided, it can no longer become another type of cell.

Different types of stem cells have different levels of potential. A stem cell that can become every type of cell in the body is called pluripotent and a stem cell that can become only some types of cells is called multipotent.

Stem cells are different from other cells in the body in three main ways:

1. Stem cells are unspecialised. They have not developed into cells that perform a specific function.
2. Stem cells can differentiate to form specialised cell types.
3. Stem cells are capable of self-renewal. Stem cells are able to divide and produce copies of themselves which leads to self-renewal.



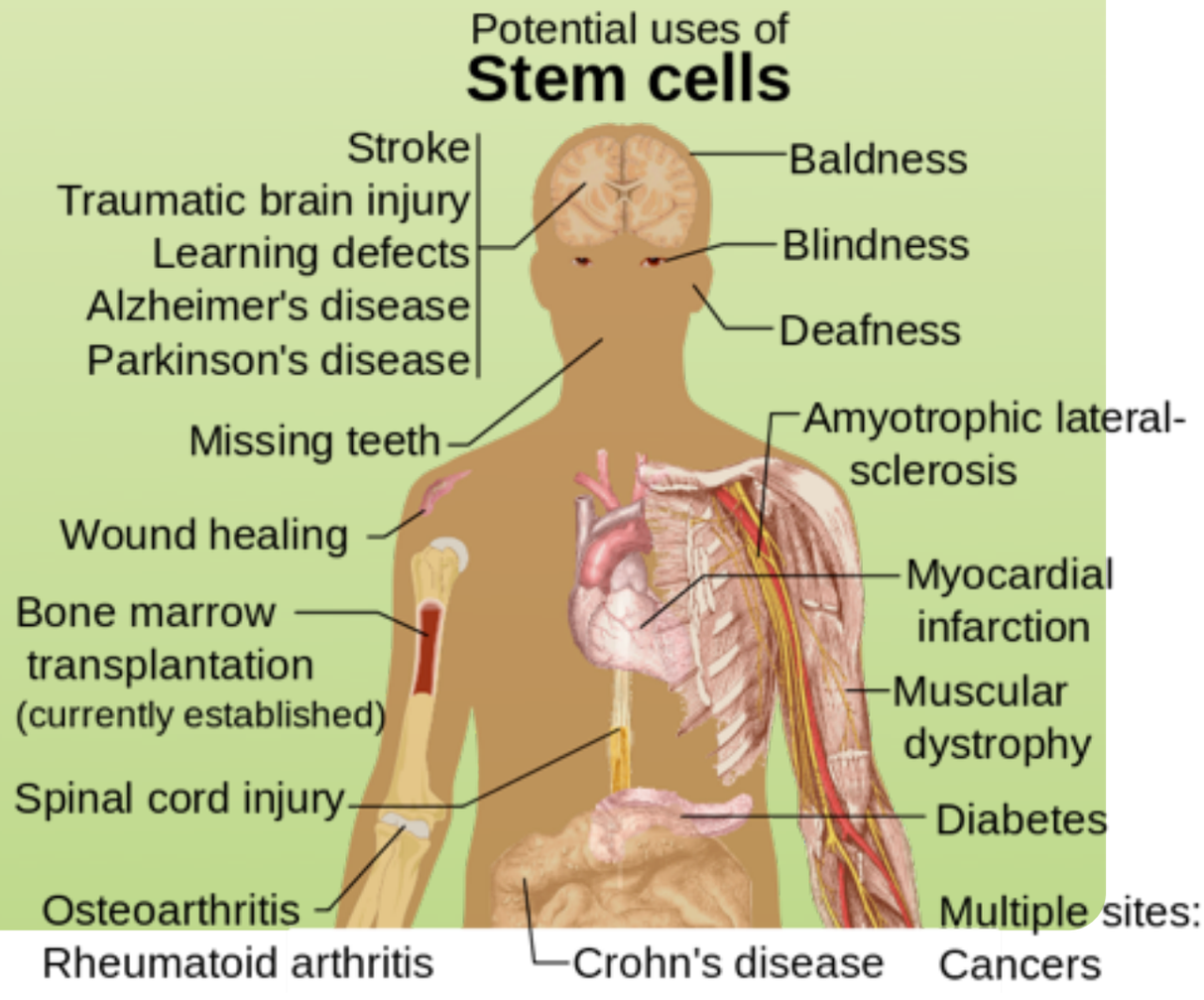
YOUR TASK

A local hospital's research team is conducting a study on degenerative diseases. They are looking for study participants who are in the early stage of these diseases and would like to try stem cell therapy. You must create an ad to attract participants.

Keep in mind that not everyone is an expert on stem cell research. Therefore your ad must include at least the following information: what a stem cell is, where stem cells come from and how stem cells might be used in treating degenerative diseases.

You should present your ad in a creative way such as a poster, slideshow, movie, comic strip, picture story book, etc.

Also ensure you include a bibliography.



Section 5

How are new cells formed?

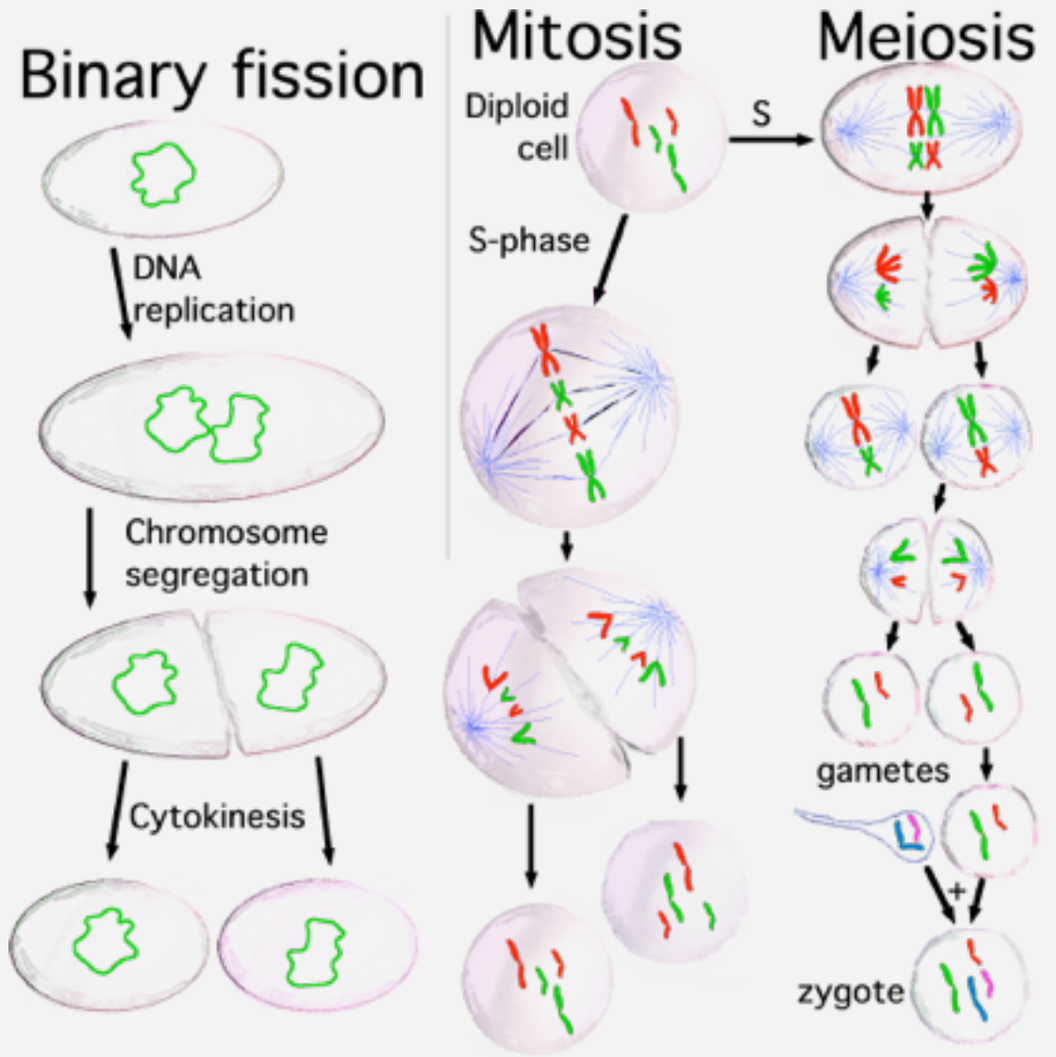
Like all living things, cells die. The number of cells that a human adult male loses per minute is roughly 96 million! Fortunately, in that same minute, about 96 million cells divided, replacing those that died. Just as you shed dead skin cells, dead cells from internal organs pass through and out of the body with waste products (poo and urine). The length of a cell's life can vary. For example, white blood cells live for about thirteen days, cells in the top layer of your skin live about 30 days, red blood cells live for about 120 days, and liver cells live for about 18 months.

The Cell Theory states that "all organisms are made up of one or more cells and that all cells come from pre-existing cells." So every time a new cell is created, it comes from a pre-existing one. The process of one cell creating another cell is known as **cell division**.

There are three main types of cell division:

- **Binary fission** occurs in prokaryotes when they replicate or duplicate. It is a form of **asexual reproduction.**
- **Mitosis** is also a form of cell replication. It occurs in eukaryotes to produce more cells for growth or to repair or replace dead cells.

Image 1.31 The three types of cell division

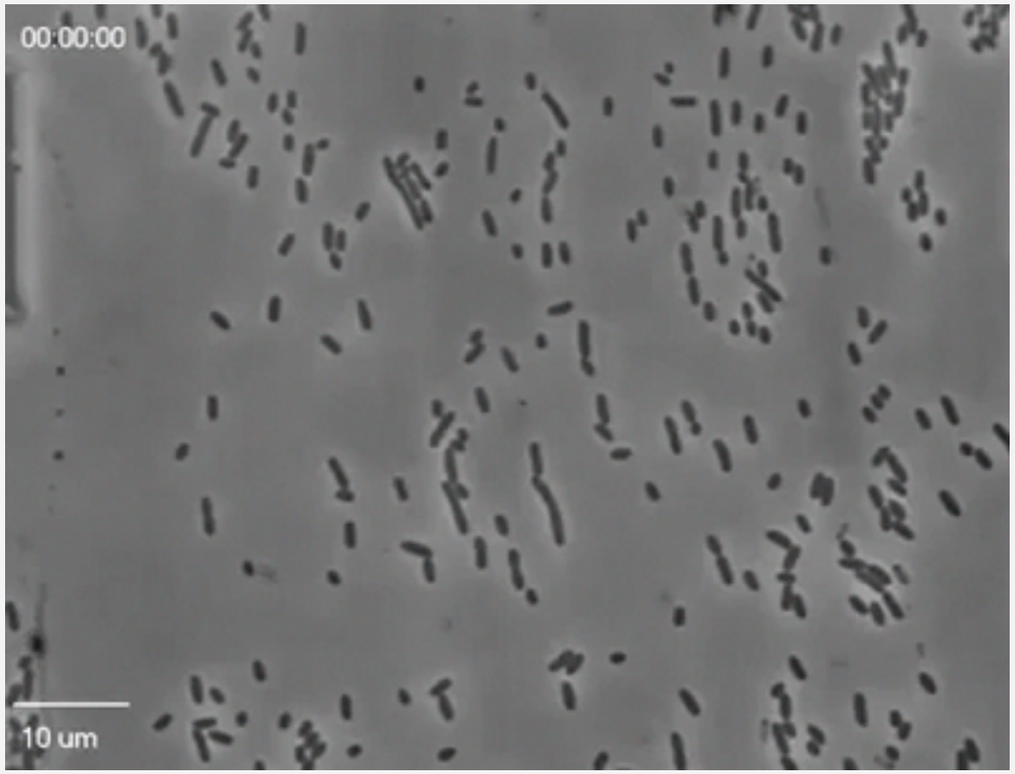


- **Meiosis** also occurs in some eukaryotes. Meiosis forms sex cells or gametes which are subsequently required for **sexual reproduction**.

BINARY FISSION

In binary fission, the prokaryotic cell starts by duplicating its DNA to create two complete sets. It then grows to a size much larger than it usually is. As the cell grows, the sets of DNA move to opposite ends of the cell. Once the cell has achieved the right size, it splits in two, creating two identical daughter cells with the exact same DNA. The process of splitting in two is called **cytokinesis**. After binary fission, each daughter cell is a clone of the parent cell.

Movie 1.12 Binary fission in action



This movie illustrates a bacteria called *Escherichia coli* reproducing and dramatically increasing in numbers by binary fission.

Image 1.32 Cytokinesis



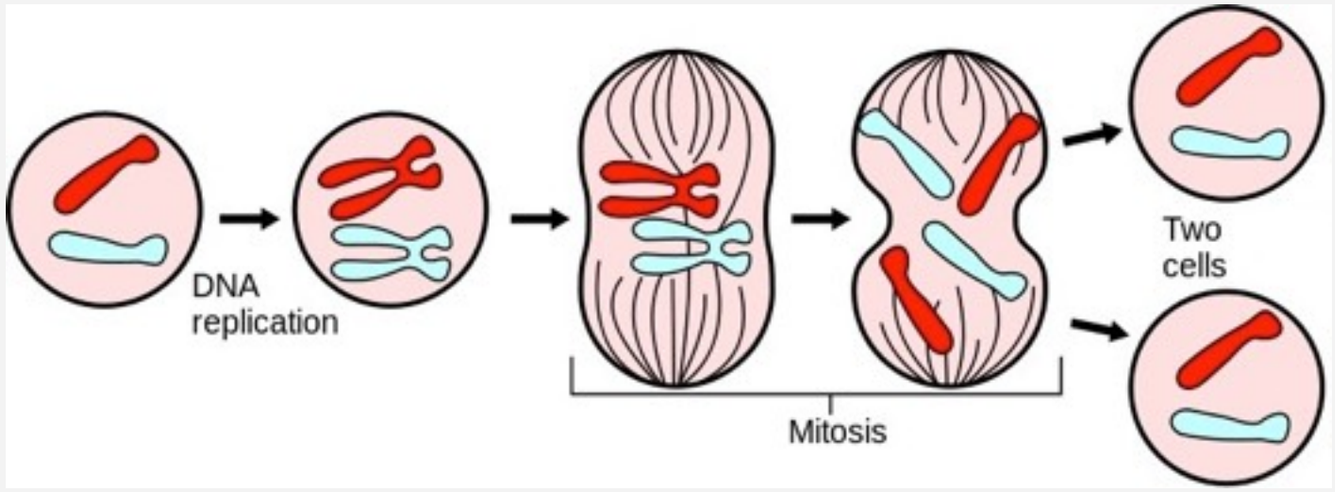
A single celled protist undergoing cytokinesis to form two protists.

When conditions are favourable, such as the right temperature and enough nutrients are available, some bacteria like *Escherichia coli* can divide every 20 minutes. This means that in just 7 hours one bacterium can generate 2,097,152 bacteria. After one more hour the number of bacteria will have risen to a colossal 16,777,216. That's why we can quickly become ill when harmful bacteria invade our bodies.

MITOSIS

Mitosis also starts with a cell duplicating its DNA to create two complete sets. The DNA then condenses to form special structures called **chromosomes**. The chromosomes are then

Image 1.33 Mitosis



Mitosis results in the formation of two cells that are genetically identical to the cell that started.

separated and move to opposite ends of the cell. It then splits in two via cytokinesis to create two daughter cells with identical DNA.

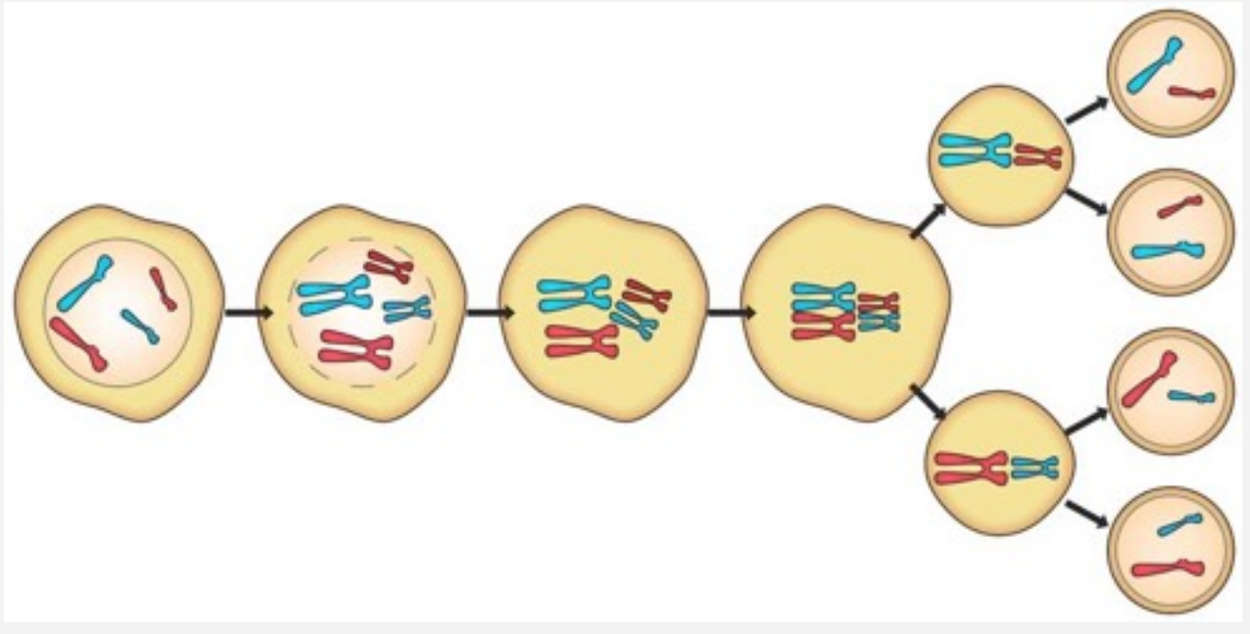
Mitosis is essential so that a multicellular organism (like you) can:
- grow.
- replace dead or injured cells.
- renew parts of the body that are shed like the skin.
- to make more cells if they are needed for a specific function.

MEIOSIS

Once again meiosis begins with a cell duplicating its DNA. However, in meiosis the parent cell then splits not once but twice. This results in the formation of four daughter cells with each having half the amount of regular DNA. The daughter cells formed after meiosis are called **gametes** or reproductive cells.

Regular human cells have 23 pairs of chromosomes, or 46 total chromosomes. Human gametes on the other hand only have 23 chromosomes in total. This is important as when a female gamete (or egg) with 23 chromosomes combines with a male gamete (or sperm) with 23 chromosomes the newly formed organism will once again have a total of 23 pairs, or 46 total chromosomes - just like all other regular human cells. When the genetic

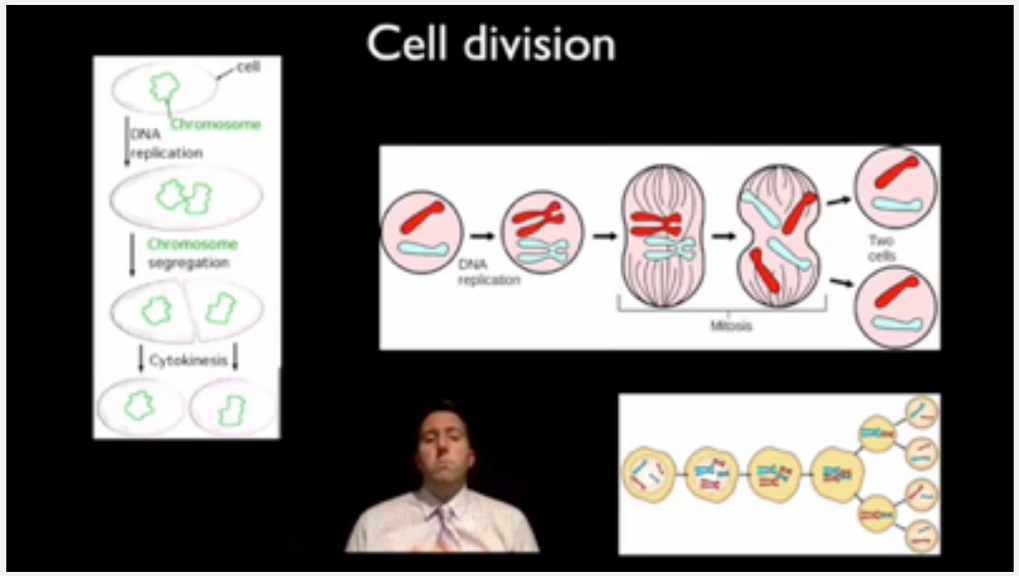
Image 1.34 Meiosis



Meiosis results in four daughter cells, each with half of the DNA of the parent cell.

information from two individuals combine in this way it is known as **sexual reproduction**.

Movie 1.13 How do cells make more cells?



QUESTIONS

1. Explain why cell division is important for an organisms or a species' survival.
2. Create a venn diagram to compare any *two* of: binary fission, mitosis or meiosis.
3. If one *Escherichia coli* bacterium replicates every 20 minutes create a table and a graph to illustrate the number of *Escherichia coli* bacteria there would be after 5 hours.
4. Explain what would happen to your body if mitosis was permanently switched off.

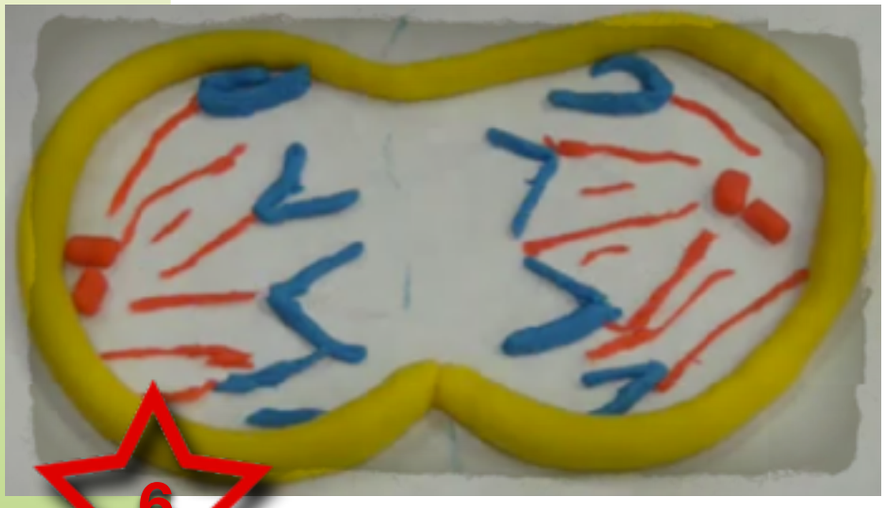
Using dot points, summarise the contents of this section.

PROJECT - Explaining Cell Division

As cell division is an actual physical process that can be seen under a microscope, when explaining it to others it can be good to show them exactly how it happens.

Your task is to create a stop motion movie to illustrate one of the three types of cell division.

You will need:
coloured plasticine, pipe cleaners, coloured paper, textas, camera, stop motion app.



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EXPERIMENT: EXPLORING MITOSIS

[To be performed under teacher direction in a science laboratory]

Aim: To observe cells undergoing mitosis and discover the four stages of mitosis.

Materials: Pre-prepared microscope slides of onion root tip cells, light microscope.

Method:

1. Pick one of the pre-prepared microscope slides and get it into focus using the microscope.
2. Using pencil, draw a diagram of what you see. Make sure your diagram includes clear and accurate labels (and is large enough to be easy to understand). Diagrams should also have a title with the specimen name and magnification used.



Results:

Draw a labelled diagram of what you observed under a microscope.

Discussion:

- Describe what you observed during this experiment.
- Find out what the four stages of mitosis are and try to include these details on your diagram.
- Discuss anything you could have done better when performing this experiment.

Conclusion:

Sum up your experiment with reference to the aim.

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Section 6

What is cancer?

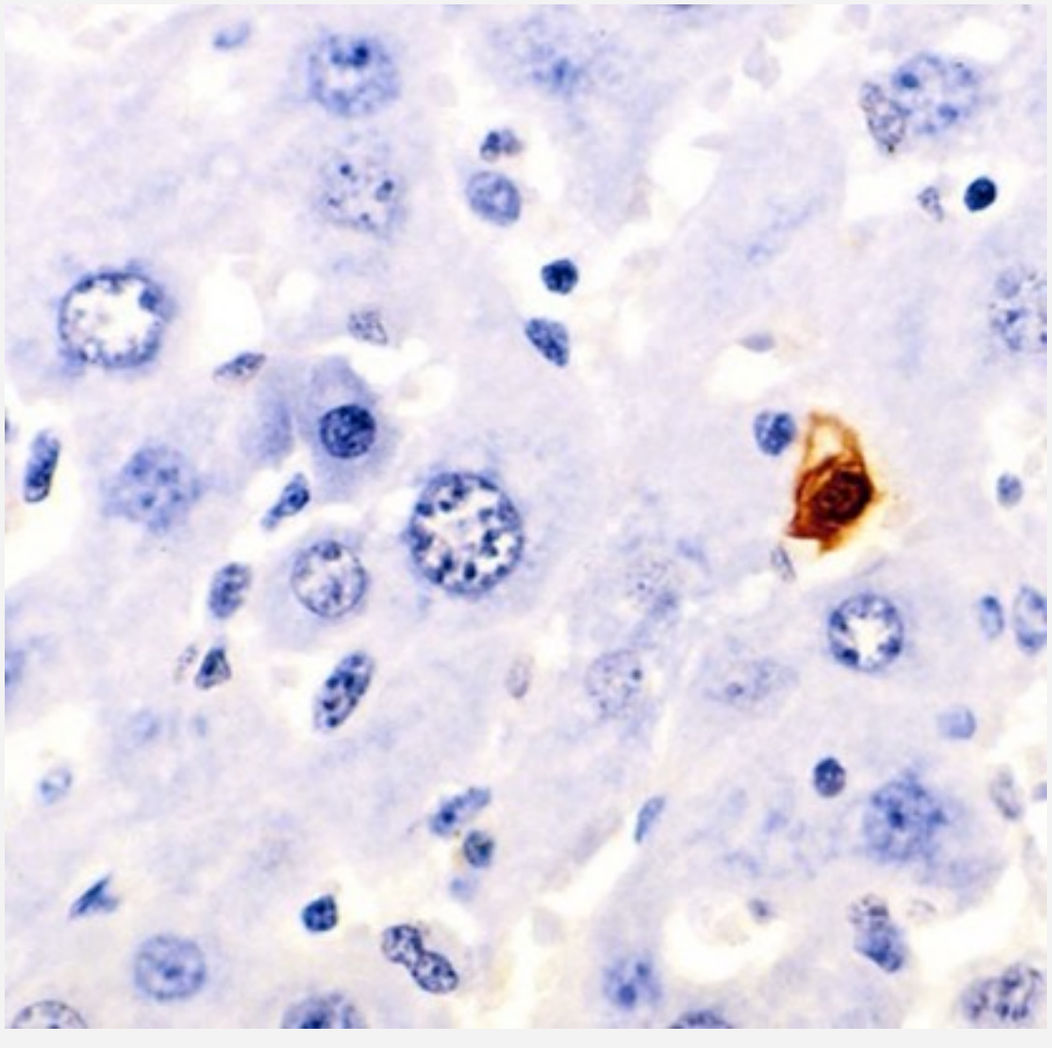
Cells in the human body are constantly creating new cells via **mitosis**. When we are young this process enables us to grow bigger - human adults are significantly bigger than human babies!

However, the process of mitosis continues into adulthood. So why do humans not get bigger and bigger throughout their entire life? The answer is that, as new cells are constantly created, old cells are constantly being killed and destroyed.

When cells die as a normal and controlled part of an organism's development it is called **apoptosis**. Apoptosis is sometimes referred to as programmed cell death, and indeed, the process of apoptosis is controlled by the cell to ensure the organism remains healthy.

Apoptosis is referred to as programmed cell death because it is controlled by the instructions (or genes) found in the DNA of a cell's nucleus. Our 40 trillion cells are constantly dying via apoptosis and constantly being replaced via mitosis.

Image 1.35 Apoptosis in action



A section of mouse liver showing a cell (stained orange) undergoing apoptosis.

However, very, very rarely the DNA instructions in a cell become damaged so that they do not work any more. Mostly when this happens the cell simply stops functioning and dies.

Although sometimes, when precisely the right combination of DNA instructions are damaged, the cells ability to self-destruct via apoptosis is destroyed. When a

single cell is not able to undergo apoptosis but is still able to undergo mitosis the cell will continue to replicate, and replicate, and replicate, and replicate. The single damaged cell will become two, then four, then eight, then sixteen, then thirty-two and so on.

This cluster of cells which are replicating out of control and not dying is called a **tumour**. When an individual has one or more tumours they are said to have **cancer**.

Tumours are considered either benign or malignant.

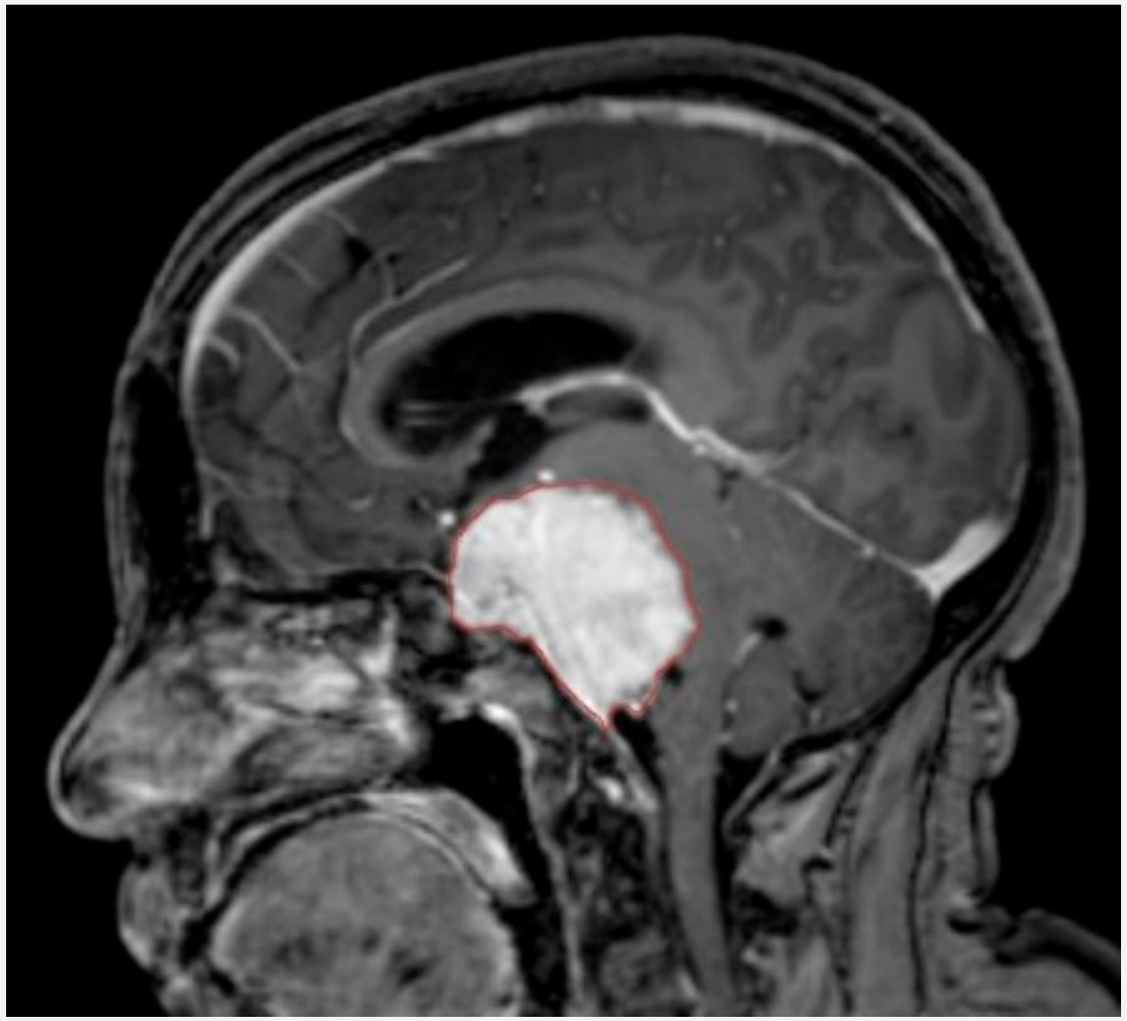
Benign tumours are rarely life threatening and do not spread to other parts of the body. They can often be removed.

Malignant tumours often invade nearby tissue and organs, spreading the disease.

The cells within malignant tumours have the ability to invade neighbouring tissues and organs, thus spreading the disease. It is also possible for cancerous cells to break free from the tumour site and enter the bloodstream, spreading the disease to other organs. This process of spreading is called **metastasis**.

Cancer can be life threatening. The key is in early diagnosis. If cancer is diagnosed early, there is a high chance that the cancer can be completely cured.

Image 1.36 A benign brain tumour called a meningioma



The brain is wrapped in layers of membranes called meninges. A meningioma is a tumour that grows out of the meninges.

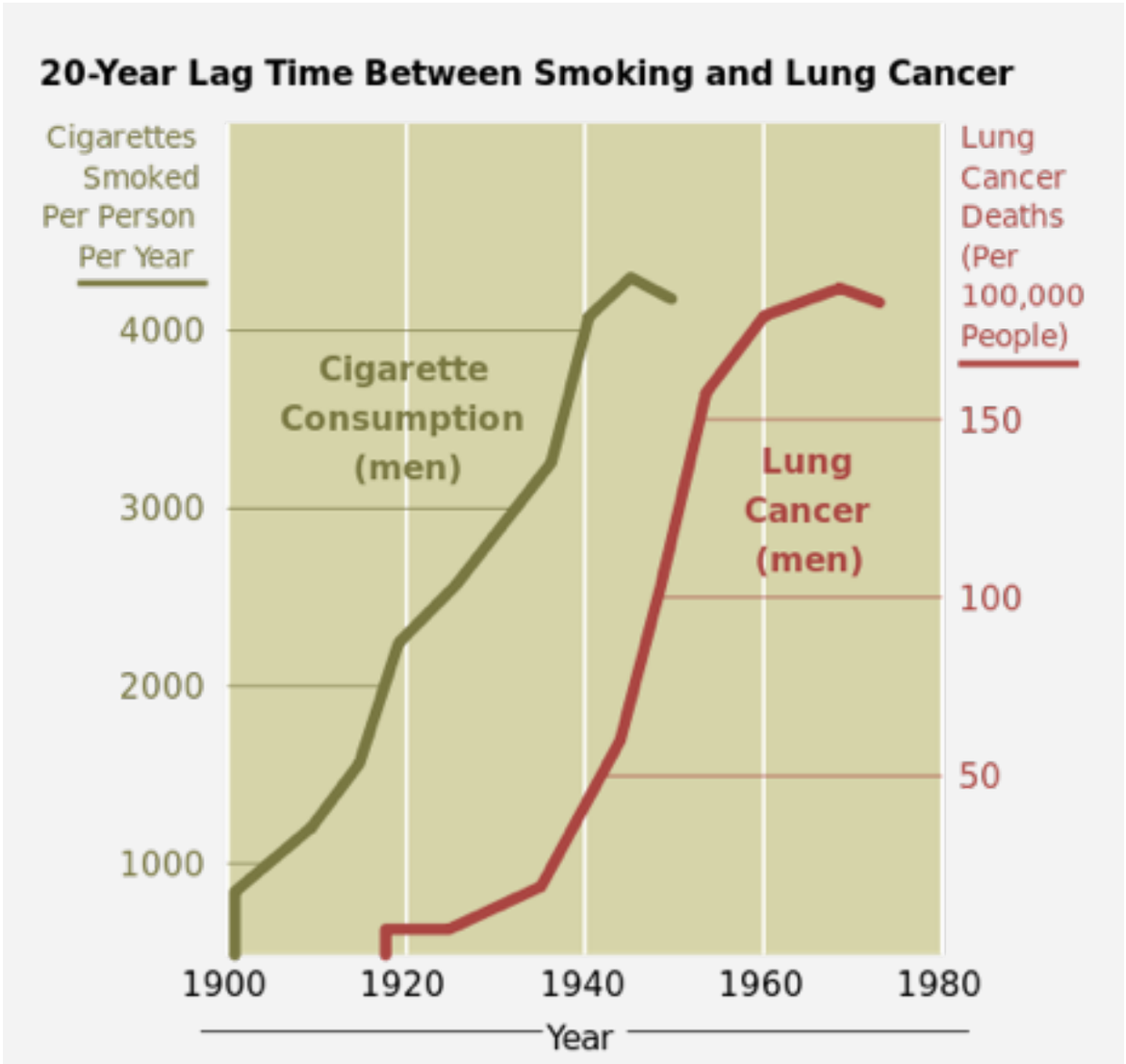
The treatment of cancer often involves more than one therapy. This may include the following:

•Chemotherapy to kill cancer cells or shrink tumours using strong medicine. This treatment usually requires several sessions and has side effects. These can include tiredness, hair loss, nausea, and vomiting. But medications are available to help treat certain side effects.

•Surgery to remove all or part of a tumour.

•Radiation to kill cancer cells or shrink tumours using high-energy waves.

We all have a risk of developing cancer. Many cancers seem to develop for no apparent reason. However, certain risk factors are known to greatly increase the chance that one or more of your cells will become abnormal and lead to cancer.



Cigarette smoking is a known cause of lung cancer.

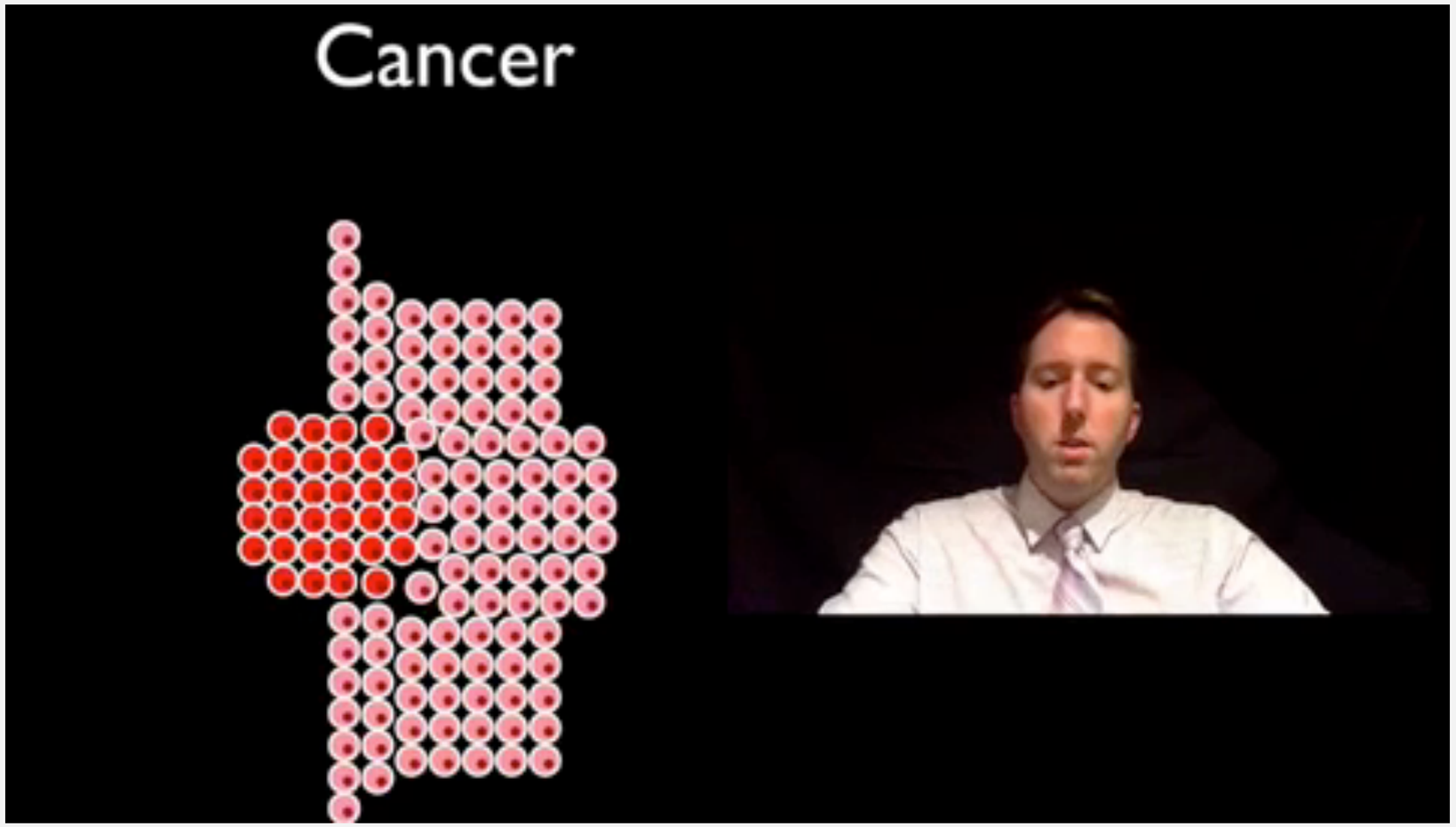
Your risk of cancer can increase through exposure to cancer causing agents such as:

- Smoking tobacco and passive smoking.
- Deliberate high exposure to sunlight.
- Excessive exposure to some chemicals such as asbestos or formaldehyde.

QUESTIONS

1. Find out what the p53 protein is and what it has to do with preventing cancer.
2. Create a venn diagram to illustrate the differences between benign and malignant tumours.
3. Create a venn diagram to illustrate the differences between mitosis and apoptosis.
4. Research another cancer causing agent or lifestyle choice that increases the likelihood of cancer. Make sure it is one not already listed in this book.
5. What is one possible change that you or someone you know could make to their lives to reduce their risk of getting cancer.

Movie 1.14 What does mitosis and apoptosis have to do with cancer?



Using dot points, summarise the contents of this section.

PROJECT - HeLa who?

Imagine you work on the ethics board of a large scientific institution.

You become aware that the cells of a woman named Henrietta Lacks are being widely used by scientists worldwide without her or her families permission.

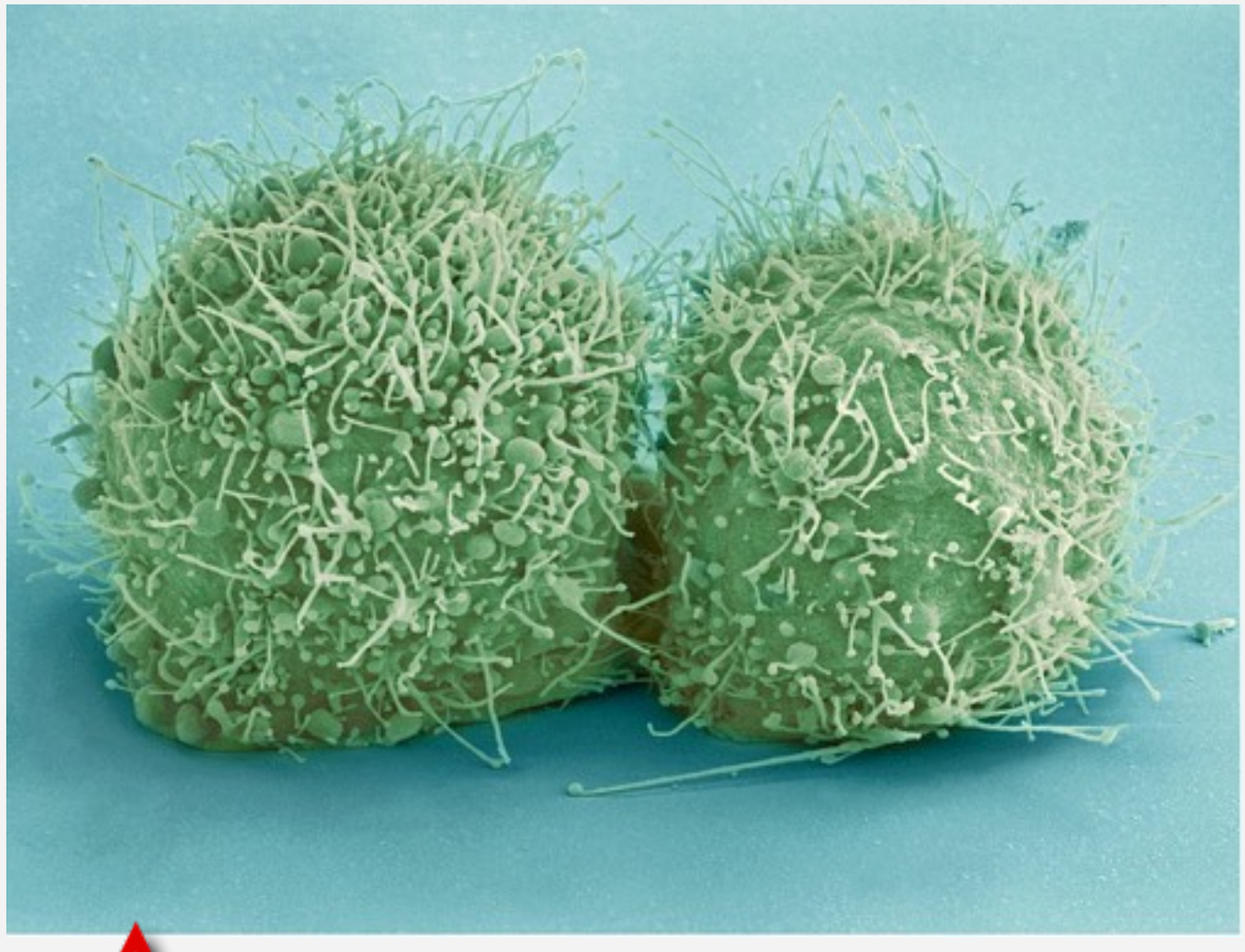
Your task is:

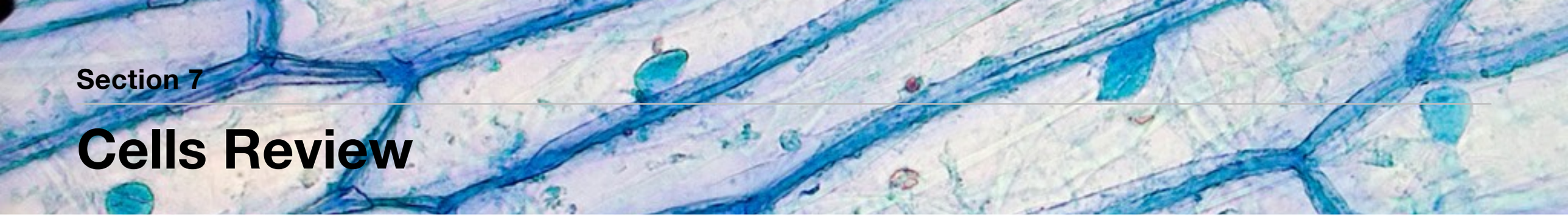
- to discover some of the major discoveries and scientific advances that have been made as a result of the use of Henrietta's cells in scientific research.
- to uncover when, where, why and how these cells were first introduced into scientific research.
- to make an informed decision about the ethical use of Henrietta Lack's cells in future research. Should the use of her cells be allowed to continue? Justify your decision.

Your work should be presented in an accurate, informative and interesting way. Think about a movie, poster, leaflet, etc.

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Image 1.37 Just-divided HeLa cells





Section 7

Cells Review

Review 1.1 Cells

Question 1 of 40

What is a microscope?

- **A.** A tiny piece of equipment.
- ✓ **B.** Something used to study small objects.
- **C.** A measuring tool.
- **D.** The machine used to take X-rays.

◀ Check Answer ▶

FURTHER ACTIVITIES

- Research some other uses of lenses in addition to microscopes. For examples, telescopes, prescription glasses, etc.
- Create a poster explaining the 5 kingdoms of life, or the three domains of life (or a variant of the six kingdoms of life).
- Find out how stem cells are currently being used in scientific research around the world.
- Find out more about the scientists involved in the original discovery of cells. What city did they live in? Did they have children? Where did they do their work? Then write a day in the life diary entry for one or all of them.
- Find out what a synchrotron is and how it is helping scientists discover more about the nature of our cells and the way they work.

Thanks

I hope you have enjoyed reading, watching, listening and interacting with this book and that you have learnt a lot!

As always a special thanks to the most important people in my world ... my family.

Thank you also to public domain contributors and Wikimedia for some imagery.